CTDMS

Central Transport & Dispatch Management System

Master (Administrator) User Manual

Demonstration Edition

From a completely empty database to dispatch, delivery, bill checking, reports and control, with a companion video and transcript

Prepared for Tulsyan CFA | Application version 1.5.1 | October 2026

Every master record, invoice, gatepass and bill in this document is a DEMONSTRATION ENTRY. They are marked DM-, (Demo) or DEMO- and live in a separate demonstration database.

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1. Read this first

DEMONSTRATION Nothing in this manual is real business data. The demonstration was run on a brand-new, completely empty database, through the same screens the staff will use. No sample data was pre-loaded by a script: every master record and every transaction was typed or uploaded through the application, exactly as a first-time Master would do it, and the video shows it.

How demonstration entries are labelled

Kind of record	How it is marked	Example
Master records (companies, customers, transporters, areas, CFA, users, ERPs)	Code starts with **DM-**; names of companies, customers, transporters and users end with **(Demo)** or start with **demo.**	DM-C01 · Alpha Stockist (Demo)
Invoices (every source)	Invoice number starts with **DEMO-**; remarks say 'Demonstration'	DEMO-SAP-1001
Gatepasses	Remarks say 'Demonstration gatepass ...' and carry the DM-RPR CFA code	DS/DM-RPR/2627/00001
Transporter bills	Bill numbers are made-up; the company is demonstration	ART/2610/001
Import files and sample photo	File names start with **DEMO_**	DEMO_MARG_sales_register.csv
Screen and video	A permanent orange banner reads 'DEMONSTRATION DATA - NOT REAL'	top centre of every video frame

Why they can be removed without touching real work

- The demonstration lives in its own database file (**demo_database/ctdms_demo.db**), started on its own address (port 8002) with its own batch file. Your real database (**data/ctdms.db**) was not opened, read or changed.
- To go live, start the application normally against a fresh database: run *python -m app.cli init-db* and *create-admin*. Nothing from the demonstration comes with it.
- If somebody ever loads these records into a live database by mistake, every one is recognisable by the DM- / (Demo) / DEMO- marks. CTDMS never deletes records that have been used (it deactivates them, so history stays), so keep demonstration and live data apart.

What is delivered together with this manual

Item	File
This manual	CTDMS_Master_User_Manual_Demonstration_Edition.pdf
Live working video (screen recording with voice and on-screen captions), 25 minutes	CTDMS_Master_Walkthrough_Video.mp4
Transcript of the video (with time stamps) and subtitle file	CTDMS_Master_Walkthrough_Transcript.txt, CTDMS_Master_Walkthrough_Subtitles.srt
Files that were imported or exported during the demonstration	demo_files\ and exports\ folders
The demonstration database and a one-click starter	demo_database\ctdms_demo.db, start_demonstration.bat
Scripts that produced everything (for repeating the demonstration)	tools\

2. Summary for top management

How the system works from start to finish (229 words)

CTDMS is the single control room between your billing systems (SAP, MARG, CBO, Intrect) and the trucks on the road. **1. Invoices arrive** from each system by Excel or CSV file, by direct API push, by scanning the e-invoice QR code, or by typing. CTDMS checks every one for duplicates, unknown customers and missing details, and shows the problem instead of hiding it. **2. Freight is priced automatically** from each transporter's approved rate card: cases multiplied by the rate for the customer's area. A new rate never works until a second person approves it. **3. One gatepass** covers a whole pickup, even with several companies, customers and towns. It must be printed before goods leave; the guard scans it and confirms gate-out. **4. The transporter's receipt (LR) number** is recorded and customers can be alerted. **5. Delivery is proven**: delivered, short, damaged or refused, with the signed receipt photographed. **6. Transporter bills are audited**: every line is compared with our own gatepasses. Wrong rates, double billing, unknown LRs and goods already paid in cash are caught before a different person approves and pays. **7. Management sees** cost per case, per transporter and per area, on-time delivery and variance, and can export everything to Excel, CSV, PDF or the API. **8. Control**: every change is logged with who, when and why, and each person sees only what their role allows.

TIP What the owners gain: fewer overpaid freight bills, proof that goods reached the customer, a record of who authorised every change, and one set of figures in place of four billing systems.

3. Checklist of master records created

Every record below was created by hand through the screens, in the order shown (each needs the one before it). Tick-marks mean the record exists in the demonstration database. **Nothing else was created.**

3.1 Source systems (Administration › Integration Settings)

	Code	Name	How data arrives	Status shown
☑	DM-SAP	SAP S/4HANA - Company A (Demo)	Excel / CSV upload	Working (file)
☑	DM-MARG	MARG ERP 9+ - Company B (Demo)	MARG adapter (file meanwhile)	File only; database link not configured
☑	DM-CBO	CBO - Company C (Demo)	REST API push	API key issued
☑	DM-INTRECT	Intrect - own distribution (Demo)	Excel / CSV upload	Inactive (planned)

3.2 CFA (warehouse) and delivery areas

	Code	Name	Detail
☑	DM-RPR	Raipur CFA (Demo)	Raipur, linked to DM-SAP as default source
☑	DM-RAIPUR	Raipur	route Local
☑	DM-BILASPUR	Bilaspur	route North, 115 km
☑	DM-DURG	Durg	route West, 40 km
☑	DM-BHILAI	Bhilai	route West, 35 km
☑	DM-KORBA	Korba	route North, 200 km

3.3 Companies (principals)

	Code	Name	Source ERP	GSTIN (demonstration)
☑	DM-COA	Company A Pharma (Demo)	DM-SAP	22DEMOA0001A1Z1
☑	DM-COB	Company B Healthcare (Demo)	DM-MARG	22DEMOB0002B1Z2
☑	DM-COC	Company C Lifesciences (Demo)	DM-CBO	22DEMOC0003C1Z3

3.4 Customers

	Code	Name	Type	Area	ERP codes recorded
☑	DM-C01	Alpha Stockist (Demo)	Stockist	Raipur	DM-MARG: ALP001
☑	DM-C02	Beta Medical Agency (Demo)	Stockist	Bilaspur	DM-MARG: BET002
☑	DM-C03	Gamma Pharma Distributors (Demo)	Stockist	Durg	-
☑	DM-C04	Delta Chemists (Demo)	Retailer	Bhilai	DM-MARG: DEL004
☑	DM-C05	Epsilon City Hospital (Demo)	Hospital	Korba	-
☑	DM-C06	Zeta Medical Hall (Demo)	Retailer	Raipur	DM-MARG: ZET006
☑	DM-C07	Eta Healthcare Agency (Demo)	Stockist	Bilaspur	DM-MARG: ETA007
☑	DM-C08	Theta Pharma Mart (Demo)	Retailer	Durg	-

3.5 Transporters and outside auto

	Code	Name	Payment	Special
☑	DM-T1	Alpha Roadlines (Demo)	To be billed (bill later)	-
☑	DM-T2	Beta Cargo Movers (Demo)	To be billed (bill later)	-
☑	DM-T3	Gamma Express Parcel (Demo)	Cash at dispatch	-
☑	DM-T4	Delta Logistics (Demo)	To be billed (bill later)	Direct pickup ₹200 per trip
☑	DM-T5	Epsilon Transport Co (Demo)	To be billed (bill later)	-
☑	DM-AUTO1	Local Auto - Ramesh (Demo)	Cash	₹150 per trip

3.6 Transporter rate cards (₹ per case, effective 01-10-2026, each approved by the checker)

Transporter	Raipur	Bilaspur	Durg	Bhilai	Korba
Alpha Roadlines (Demo)	6	22	14	15	28
Beta Cargo Movers (Demo)	7	20	13	14	30
Gamma Express Parcel (Demo)	no rate	21	no rate	no rate	26
Delta Logistics (Demo)	8	24	15	16	32
Epsilon Transport Co (Demo)	5	19	no rate	13	no rate

Gamma Express serves only Korba and Bilaspur, and Epsilon Transport does not serve Durg or Korba. This is deliberate: it shows the 'no rate' warning at the gatepass.

3.7 Users (Administration › Users)

	Username	Name	Role	Why it is needed
☑	admin	Master Administrator	Administrator	The Master: created from the command line, as the first user always is
☑	demo.checker	Demo Checker	Checker / Approver	Approves rates and bills (two-person rule)
☑	demo.operator	Demo Dispatch Operator	Dispatch Operator	Warehouse desk; sees only Raipur CFA
☑	demo.accounts	Demo Accounts	Accounts	Types and pays transporter bills

3.8 Settings changed by the Master

	Setting	Value	Reason
☑	org.name / org.address	Tulsyan CFA (Demonstration) / Raipur ...	Printed on gatepasses and reports
☑	notify.provider	LOG (preview)	Customer alerts are prepared and shown, nothing is sent; WhatsApp / SMS are paid and locked
	All other settings	left at their default values	Defaults are safe: approvals ON, printed gatepass required

4. All information given and used as samples

Everything below was given to the system during the demonstration. Dates are 14 to 17 October 2026.

4.1 Files that were uploaded

File (folder demo_files)	Source system	Format	Content	What it demonstrates
DEMO_SAP_billing_documents_itemwise.xlsx	SAP, Company A	Excel, one row per medicine	5 invoices, 13 medicine lines, 6 SKUs	Column auto-detection; SKU detail; customer found by central code
DEMO_MARG_sales_register.csv	MARG, Company B	CSV, one row per invoice	7 rows: 6 invoices + 1 duplicate	Unknown party code mapped once; blank case count becomes an exception; duplicate refused; layout saved as a template
DEMO_CBO_invoice_export.csv	CBO, Company C	CSV	3 invoices	Customer recognised by GST number when the code is unknown
(typed on screen)	Manual entry	Add Invoice form	2 invoices (one cancelled)	Hand entry; cancel with a reason
DEMO_einvoice_QR_for_scanning.txt	e-invoice QR	QR text	1 invoice (4 cases counted at the desk)	Invoice created from a QR scan at the dispatch desk
(API call, key issued to CBO)	CBO API push	JSON	1 invoice DEMO-CBO-3101	Another system writing into CTDMS with no file
DEMO_rate_card_DM-T1 … DM-T4.csv	Transporters	CSV (Area, Rate, Effective from)	4 rate lists, 20 rates	Bulk rate upload with approval
(typed on screen)	Epsilon Transport	Rate form	3 rates	Typing rates one by one
DEMO_transporter_bill_Beta_Cargo.csv	Beta Cargo Movers	CSV statement	5 bill lines with 3 deliberate errors	Rate too high, same LR billed twice, LR not in our records
DEMO_signed_LR_photo.png	Customer receipt	Image	Sample photo of a signed LR	Proof of delivery attachment

4.2 Invoices as they stand at the end of the demonstration

Invoice	Source · company	Customer · area	Cases · value	Status now	Gatepass · carrier · LR	Freight	Delivery
DEMO-CBO-3001	DM-CBO · COC	Gamma Pharma Distributors (Demo) · Durg	10 · ₹41,000	Gone out	G2 · Beta Cargo Movers (Demo) · LR BCM-7002	₹130	Delivered short
DEMO-CBO-3002	DM-CBO · COC	Epsilon City Hospital (Demo) · Korba	16 · ₹64,500	Gone out	G3 · Gamma Express Parcel (Demo) · LR GEP-3301	₹416	-
DEMO-CBO-3003	DM-CBO · COC	Theta Pharma Mart (Demo) · Durg	4 · ₹12,800	Gatepass printed	G5 · Delta Logistics (Demo) · LR -	₹60	-
DEMO-CBO-3101	DM-CBO · COC	Eta Healthcare Agency (Demo) · Bilaspur	6 · ₹22,000	Waiting	- · - · LR -	-	-
DEMO-MAN-4001	Typed · COA	Delta Chemists (Demo) · Bhilai	6 · ₹19,200	Gone out	G1 · Alpha Roadlines (Demo) · LR ATR-1103	₹90	Delivered
DEMO-MAN-4002	Typed · COB	Zeta Medical Hall (Demo) · Raipur	2 · ₹5,000	Cancelled	- · - · LR -	-	-
DEMO-MRG-2001	DM-MARG · COB	Alpha Stockist (Demo) · Raipur	6 · ₹22,400	Gone out	G1 · Alpha Roadlines (Demo) · LR ATR-1102	₹36	Delivered
DEMO-MRG-2002	DM-MARG · COB	Delta Chemists (Demo) · Bhilai	5 · ₹18,900	Gone out	G1 · Alpha Roadlines (Demo) · LR ATR-1103	₹75	Delivered
DEMO-MRG-2003	DM-MARG · COB	Zeta Medical Hall (Demo) · Raipur	3 · ₹9,600	Gone out	G4 · Outside auto · LR -	₹75	-
DEMO-MRG-2004	DM-MARG · COB	Eta Healthcare Agency (Demo) · Bilaspur	8 · ₹31,750	Gatepass made, not printed	G7 · Epsilon Transport Co (Demo) · LR -	₹152	-
DEMO-MRG-2005	DM-MARG · COB	Theta Pharma Mart (Demo) · Durg	4 · ₹15,000	Gone out	G2 · Beta Cargo Movers (Demo) · LR BCM-7003	₹52	-
DEMO-MRG-2006	DM-MARG · COB	Beta Medical Agency (Demo) · Bilaspur	7 · ₹27,300	Waiting	- · - · LR -	-	-
DEMO-QR-5001	DM-SAP · COA	Zeta Medical Hall (Demo) · Raipur	4 · ₹14,400	Gone out	G1 · Alpha Roadlines (Demo) · LR ATR-1101	₹24	Delivered
DEMO-SAP-1001	DM-SAP · COA	Alpha Stockist (Demo) · Raipur	12 · ₹48,500	Gone out	G1 · Alpha Roadlines (Demo) · LR ATR-1102	₹72	Delivered
DEMO-SAP-1002	DM-SAP · COA	Beta Medical Agency (Demo) · Bilaspur	9 · ₹36,240	Gone out	G2 · Beta Cargo Movers (Demo) · LR BCM-7001	₹180	-
DEMO-SAP-1003	DM-SAP · COA	Gamma Pharma Distributors (Demo) · Durg	14 · ₹52,800	Gone out	G2 · Beta Cargo Movers (Demo) · LR BCM-7002	₹182	Delivered short
DEMO-SAP-1004	DM-SAP · COA	Epsilon City Hospital (Demo) · Korba	18 · ₹71,300	Gone out	G3 · Gamma Express Parcel (Demo) · LR GEP-3301	₹468	-
DEMO-SAP-1005	DM-SAP · COA	Zeta Medical Hall (Demo) · Raipur	3 · ₹8,750	Gone out	G4 · Outside auto · LR -	₹75	-

4.3 Gatepasses

Ref	Gatepass number	Carrier	Mode	Load	Freight charged	Paid at gate	State	What it shows
G1	DS/DM-RPR/2627/00001	Alpha Roadlines (Demo)	TRANSPORTER	5 inv · 33 cases	₹297	₹0	Gone out	Mixed: 2 companies (A and B), 3 customers, 2 towns (Raipur, Bhilai), 5 invoices from SAP, MARG, QR scan and hand entry. Carrier: Alpha Roadlines, to be billed.
G2	DS/DM-RPR/2627/00002	Beta Cargo Movers (Demo)	TRANSPORTER	4 inv · 37 cases	₹544	₹0	Gone out	Mixed: 3 companies (A, B and C), 3 customers, 2 towns (Bilaspur, Durg), 4 invoices. Carrier: Beta Cargo Movers, to be billed.
G3	DS/DM-RPR/2627/00003	Gamma Express Parcel (Demo)	TRANSPORTER	2 inv · 34 cases	₹884	₹884	Gone out	2 companies (A and C), 1 hospital in Korba. Carrier: Gamma Express, CASH, paid at the gate.
G4	DS/DM-RPR/2627/00004	Local Auto - Ramesh (Demo)	OUTSIDE_AUTO	2 inv · 6 cases	₹150	₹150	Gone out	Local delivery in Raipur: our own hired auto, ₹150 per trip, 2 invoices of 2 companies to the same customer.
G5	DS/DM-RPR/2627/00005	Delta Logistics (Demo)	DIRECT	1 inv · 4 cases	₹260	₹0	Printed	Direct pickup by Delta Logistics (₹200 extra per trip). Printed, goods not yet gone out.
G6	DS/DM-RPR/2627/00006	Delta Logistics (Demo)	TRANSPORTER	1 inv · 8 cases	₹192	₹0	Void	Voided: made on the wrong carrier (Delta) and cancelled with a reason before gate-out.
G7	DS/DM-RPR/2627/00007	Epsilon Transport Co (Demo)	TRANSPORTER	1 inv · 8 cases	₹152	₹0	Not printed	Replacement for the voided one: Epsilon Transport, generated but not printed yet.

4.4 Transporter bills

Bill	Transporter	Lines	Billed	Our freight	Variance	Status	History
ART/2610/001	Alpha Roadlines (Demo)	3	₹297	₹297	₹0	Paid	Typed by demo.accounts; approved by demo.checker; UTR UTR-DEMO-000123456789
BCM/2610/007	Beta Cargo Movers (Demo)	5	₹1,132	₹544	₹588	Variance	Entered by demo.accounts; blocked, not approvable

4.5 Other information recorded

Item	Content
Customer alerts (preview mode)	15 messages prepared: 'dispatched' and 'LR number' for each customer of each gatepass; marked sent, nothing transmitted
Proof of delivery	Alpha Roadlines' consignments delivered 17-10-2026; Gamma Pharma delivered short (22 of 24 cases) with a note and a photo; the others still pending
Override	DEMO-MRG-2006: case count entered as 7 by the Master with a reason (audited); invoice stays pending
Cancellation	DEMO-MAN-4002 cancelled with the reason 'Typed by mistake'; stays in the register
Audit trail	257 audit entries written automatically during the demonstration
E-mailed report	Weekly Excel gatepass register for management@example.com (needs the mail server to send)
Users	4 users: admin, demo.checker, demo.operator, demo.accounts; the demonstration password is in the appendix

5. Every invoice and bill explained in plain language

For each demonstration entry: what it represents, and how it touches freight cost, dispatch, the customer, delivery, the transporter's bill and management reports.

DEMO-CBO-3001 Gamma Pharma Distributors (Demo) · Durg · 10 cases · ₹41,000

What it represents: Company C's invoice to the Durg stockist.
How it affects the business: Company C's code CB-GAMMA was unknown, but the GST number identified the customer. It shares one LR with SAP-1003 and was delivered short.
Now: Gone out; gatepass G2; freight ₹130; delivery delivered short.

DEMO-CBO-3002 Epsilon City Hospital (Demo) · Korba · 16 cases · ₹64,500

What it represents: Company C's invoice to the Korba hospital, 16 cases.
How it affects the business: On the same cash gatepass as SAP-1004, so the hospital receives both companies' goods in one consignment.
Now: Gone out; gatepass G3; freight ₹416; delivery -.

DEMO-CBO-3003 Theta Pharma Mart (Demo) · Durg · 4 cases · ₹12,800

What it represents: Company C's invoice to a Durg retailer, 4 cases.
How it affects the business: Delta Logistics collects it from our warehouse, so a direct pickup charge of ₹200 is added. The gatepass is printed but the goods have not left, so the invoice is still not dispatched.
Now: Gatepass printed; gatepass G5; freight ₹60; delivery -.

DEMO-CBO-3101 Eta Healthcare Agency (Demo) · Bilaspur · 6 cases · ₹22,000

What it represents: An invoice pushed by Company C's system through the API.
How it affects the business: No file and no typing. It simply arrived, was validated, and is waiting for dispatch.
Now: Waiting; gatepass no gatepass yet; freight not yet calculated; delivery -.

DEMO-MAN-4001 Delta Chemists (Demo) · Bhilai · 6 cases · ₹19,200

What it represents: An invoice typed by hand for a Bhilai retailer.
How it affects the business: Useful when no export exists. It went out on gatepass 1 like any other.
Now: Gone out; gatepass G1; freight ₹90; delivery delivered.

DEMO-MAN-4002 Zeta Medical Hall (Demo) · Raipur · 2 cases · ₹5,000

What it represents: An invoice typed by mistake.
How it affects the business: Cancelled with a written reason. It stays in the register, marked cancelled, so there is a complete history, and it is excluded from totals.
Now: Cancelled; gatepass no gatepass yet; freight not yet calculated; delivery -.

DEMO-MRG-2001 Alpha Stockist (Demo) · Raipur · 6 cases · ₹22,400

What it represents: A MARG invoice from Company B to the same Raipur stockist as SAP-1001.
How it affects the business: MARG calls the customer ALP001; the code was recorded once on the customer screen, so the invoice matched without help. Same truck, same bill as SAP-1001.
Now: Gone out; gatepass G1; freight ₹36; delivery delivered.

DEMO-MRG-2002 Delta Chemists (Demo) · Bhilai · 5 cases · ₹18,900

What it represents: A MARG invoice to a Bhilai retailer, 5 cases.
How it affects the business: Bhilai freight is ₹15 a case, higher than Raipur, because of distance. On gatepass 1.
Now: Gone out; gatepass G1; freight ₹75; delivery delivered.

DEMO-MRG-2003 Zeta Medical Hall (Demo) · Raipur · 3 cases · ₹9,600

What it represents: A MARG invoice that appeared twice in the file.
How it affects the business: The second copy was refused as a duplicate, so freight and sales cannot be counted twice. The first went by the local auto.
Now: Gone out; gatepass G4; freight ₹75; delivery -.

DEMO-MRG-2004 Eta Healthcare Agency (Demo) · Bilaspur · 8 cases · ₹31,750

What it represents: A MARG invoice for a Bilaspur stockist, 8 cases.
How it affects the business: First put on the wrong carrier, which was voided before the goods left, then on Epsilon Transport. The gatepass is generated but not printed, so the goods cannot leave yet.
Now: Gatepass made, not printed; gatepass G7; freight ₹152; delivery -.

DEMO-MRG-2005 Theta Pharma Mart (Demo) · Durg · 4 cases · ₹15,000

What it represents: A MARG invoice whose party code TH-08 CTDMS did not know.
How it affects the business: The system suggested the right customer; the Master mapped the code once, and every future file will be recognised. Delivered by Beta Cargo (LR BCM-7003).
Now: Gone out; gatepass G2; freight ₹52; delivery -.

DEMO-MRG-2006 Beta Medical Agency (Demo) · Bilaspur · 7 cases · ₹27,300

What it represents: A MARG invoice that arrived without a case count, so it was held as an Exception.
How it affects the business: Without cases, freight cannot be calculated, so CTDMS refuses to dispatch it. The Master entered 7 cases with a reason (recorded). It now waits for a gatepass.
Now: Waiting; gatepass no gatepass yet; freight not yet calculated; delivery -.

DEMO-QR-5001 Zeta Medical Hall (Demo) · Raipur · 4 cases · ₹14,400

What it represents: An invoice created at the warehouse desk by scanning the e-invoice QR code printed on the paper invoice.
How it affects the business: The operator only counted the cases (4). The QR gave company, customer, number, date and value. Gatepass 1.
Now: Gone out; gatepass G1; freight ₹24; delivery delivered.

DEMO-SAP-1001 Alpha Stockist (Demo) · Raipur · 12 cases · ₹48,500

What it represents: A normal order from Company A's SAP to a Raipur stockist, 12 cases of three medicines.
How it affects the business: It shared a truck (gatepass 1) with MARG and QR invoices going to the same stockist. Freight is 12 cases at ₹6, billed by Alpha Roadlines, who were paid after the checker approved their bill.
Now: Gone out; gatepass G1; freight ₹72; delivery delivered.

DEMO-SAP-1002 Beta Medical Agency (Demo) · Bilaspur · 9 cases · ₹36,240

What it represents: An order from Company A to a Bilaspur stockist.
How it affects the business: Carried by Beta Cargo Movers. Its LR number appears on Beta's bill, where Beta charged ₹24 a case instead of the agreed ₹20. CTDMS flags the difference.
Now: Gone out; gatepass G2; freight ₹180; delivery -.

DEMO-SAP-1003 Gamma Pharma Distributors (Demo) · Durg · 14 cases · ₹52,800

What it represents: A large order from Company A to a Durg stockist, 14 cases.
How it affects the business: One LR covers it together with Company C's invoice for the same customer (24 cases). The customer received only 22, so the delivery is recorded as short. Beta's bill charges this LR twice; CTDMS blocks it.
Now: Gone out; gatepass G2; freight ₹182; delivery delivered short.

DEMO-SAP-1004 Epsilon City Hospital (Demo) · Korba · 18 cases · ₹71,300

What it represents: A hospital order, 18 cases, the biggest value in the demonstration.
How it affects the business: Sent to Korba with cash transporter Gamma Express; freight is paid at the gate, so no transporter bill is expected. Delivery proof is still awaited.
Now: Gone out; gatepass G3; freight ₹468; delivery -.

DEMO-SAP-1005 Zeta Medical Hall (Demo) · Raipur · 3 cases · ₹8,750

What it represents: A small order to a Raipur retailer, 3 cases.
How it affects the business: Delivered locally by our own hired auto together with MARG invoice 2003. The auto charge of ₹150 is for the trip, not per case, and is paid in cash.
Now: Gone out; gatepass G4; freight ₹75; delivery -.

The two transporter bills

Bill ART/2610/001 · Alpha Roadlines (Demo) · billed ₹297 · our freight ₹297 · status Paid

What it represents: Alpha Roadlines' bill for gatepass 1: three lines, ₹297 (33 cases).
How it affects the business: Every line matched our gatepass: same LR, same cases, same rate as the rate card. The variance is zero, so the checker could approve it and accounts paid it with the bank reference (UTR). Effect: freight cost is confirmed and the bill is closed; the consignments' timelines show 'bill checked' and 'freight paid'.

LR	Cases billed / ours	Rate billed / card	Amount billed	Result
ATR-1101	4 / 4	6 / 6	₹24	Matched
ATR-1102	18 / 18	6 / 6	₹108	Matched
ATR-1103	11 / 11	15 / 15	₹165	Matched

Bill BCM/2610/007 · Beta Cargo Movers (Demo) · billed ₹1,132 · our freight ₹544 · status Variance

What it represents: Beta Cargo Movers' bill for gatepass 2: the transporter says ₹1,132; our own calculation is ₹544.
How it affects the business: It has three deliberate errors: a rate of ₹24 instead of ₹20, one LR billed twice, and an LR that does not exist in our records. CTDMS flags five lines, shows the variance, and will not let the bill be approved until the lines are corrected or removed. Effect: the business avoids paying about ₹588 more than it owes.

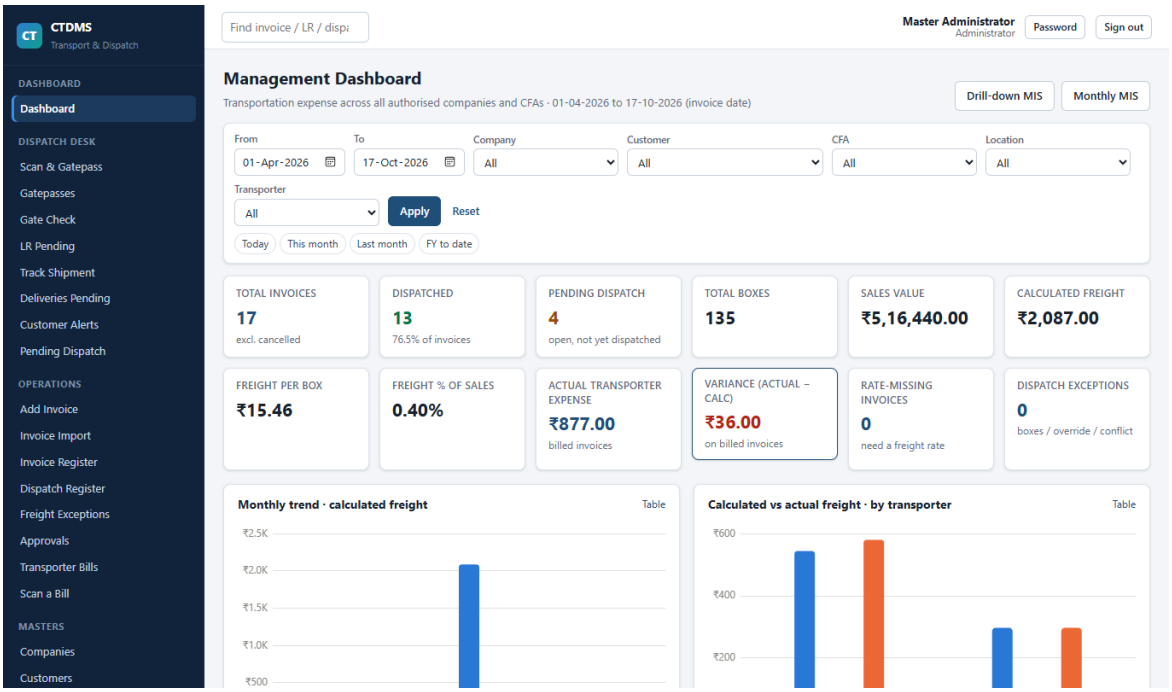
LR	Cases billed / ours	Rate billed / card	Amount billed	Result
BCM-7001	9 / 9	- / 20	₹216	Variance
BCM-7002	24 / 24	- / 13	₹312	Matched
BCM-7002	24 / 24	- / -	₹312	Duplicate Billing
BCM-7003	4 / 4	- / 13	₹52	Matched
BCM-9999	8 / -	- / -	₹240	Not Found

6. Step-by-step manual with screens

Each step follows the video. The screens are taken from the finished demonstration database.

Step 1. Sign in, change your password, read the dashboard

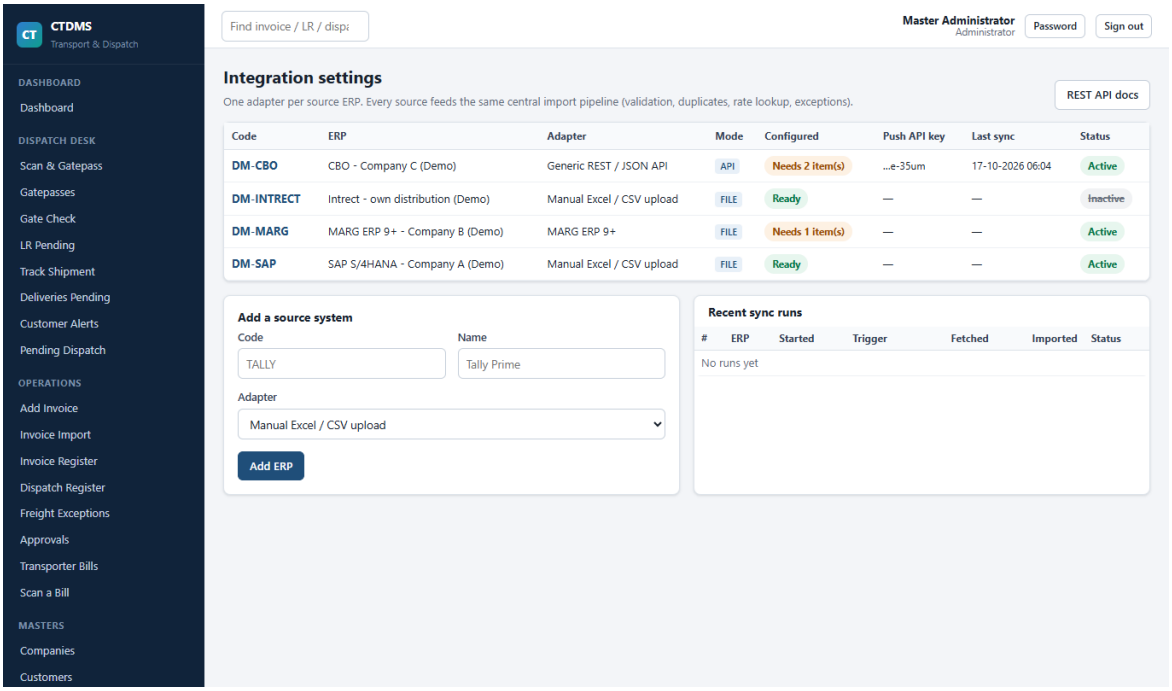
The Master signs in with the user name and password created on the command line (*create-admin*), and changes the password. Five wrong attempts lock an account for 15 minutes; the Master can unlock it from Users.



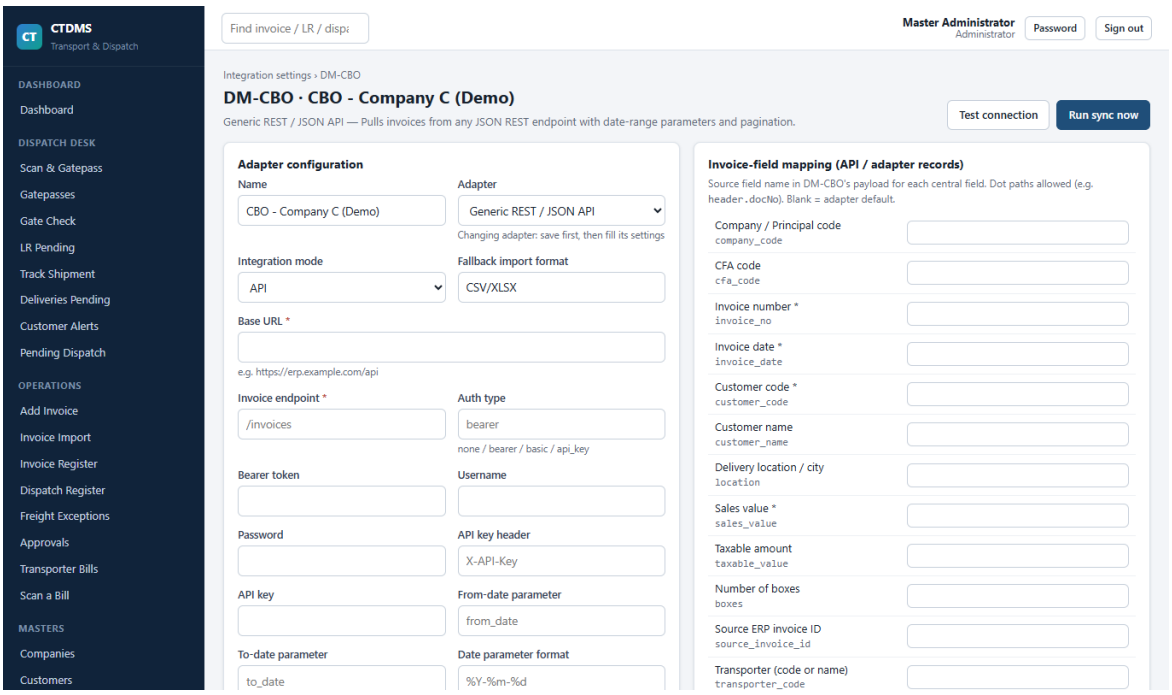
Dashboard after the demonstration: 17 invoices, 13 dispatched, freight and variance.

Step 2. Add the source systems (ERPs)

Administration › Integration Settings › Add ERP: a code, a name and an adapter. Then open each one and write its status in the notes. For CBO, *Generate API key* (shown once). For MARG, *Test connection* shows what is still missing. Untick Active to park a system.



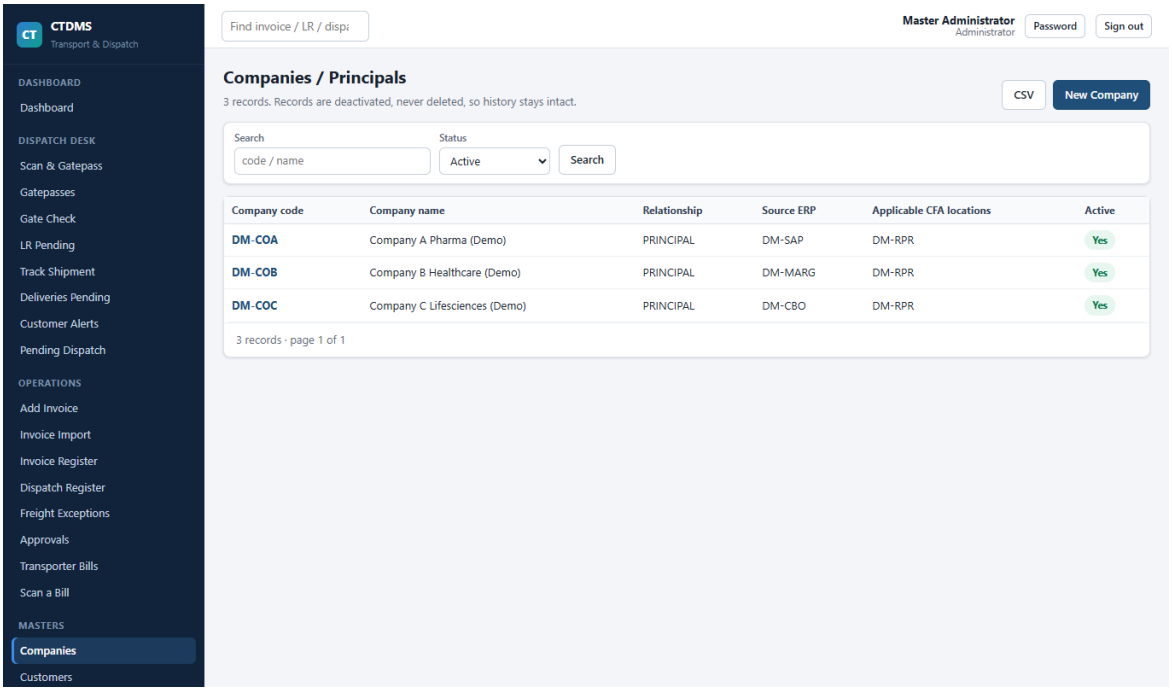
Four systems, four statuses.



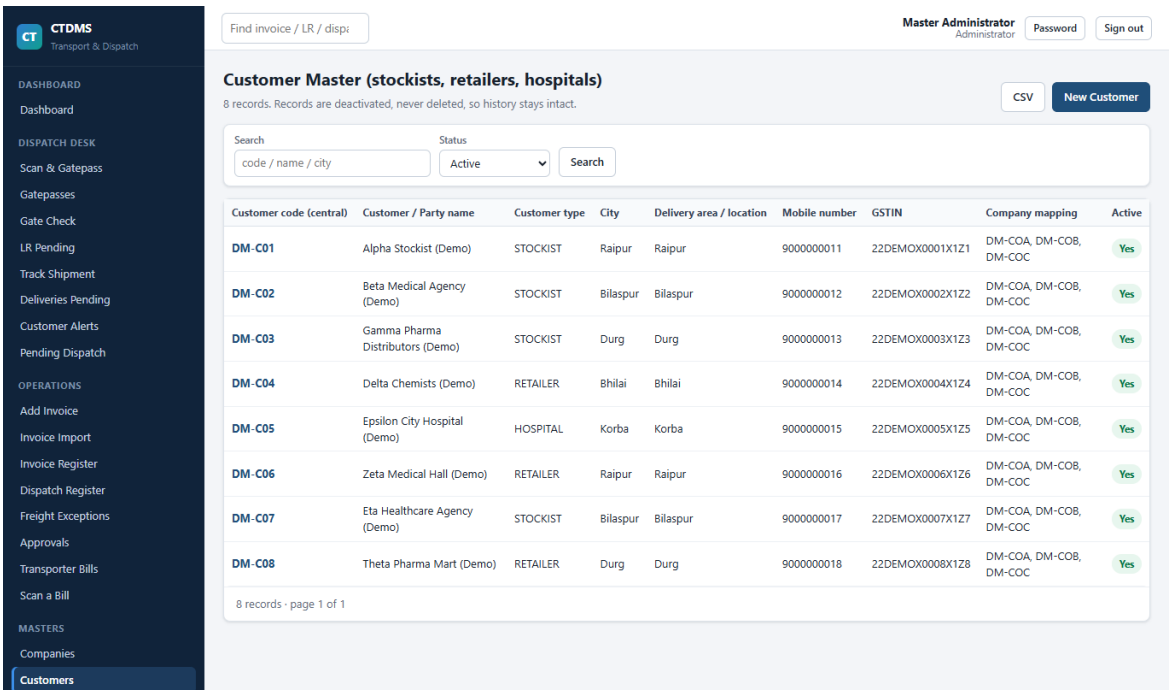
CBO: API key hint, push address, field mapping and sync history.

Step 3. Master records, in order

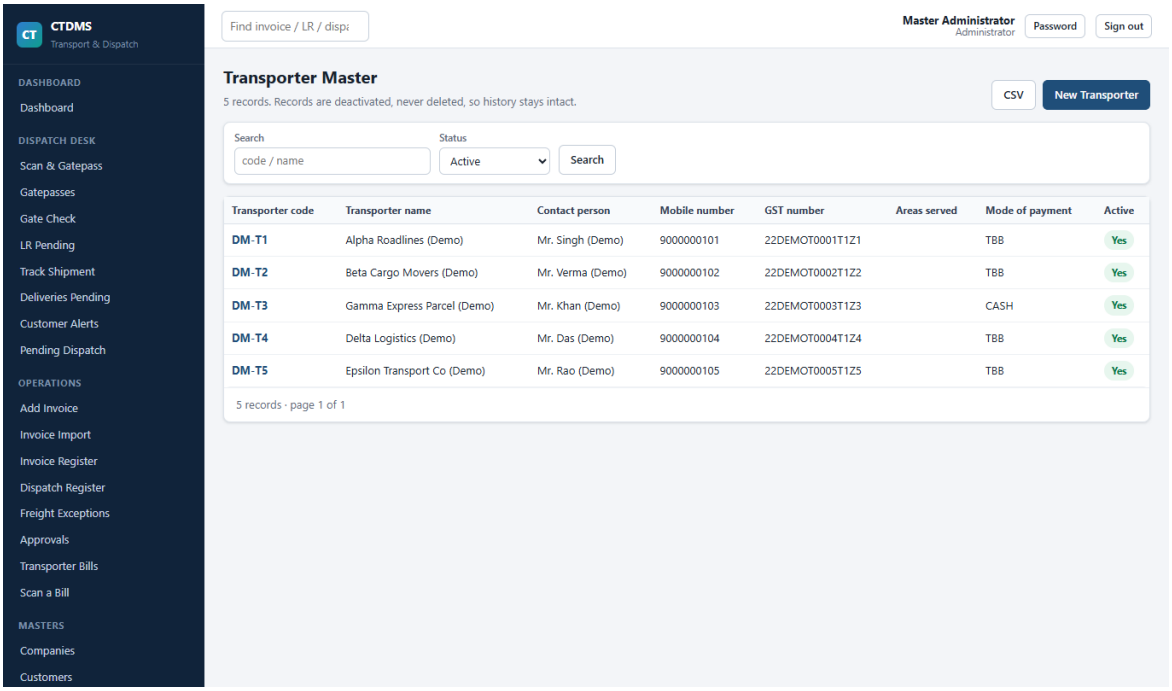
- **CFA** (the warehouse) with its default ERP; **Areas**; **Companies** (linked to their ERP and the CFA); **Transporters** (cash or to-be-billed); **Outside auto**; **Customers** (area, mobile, GST number, companies served).



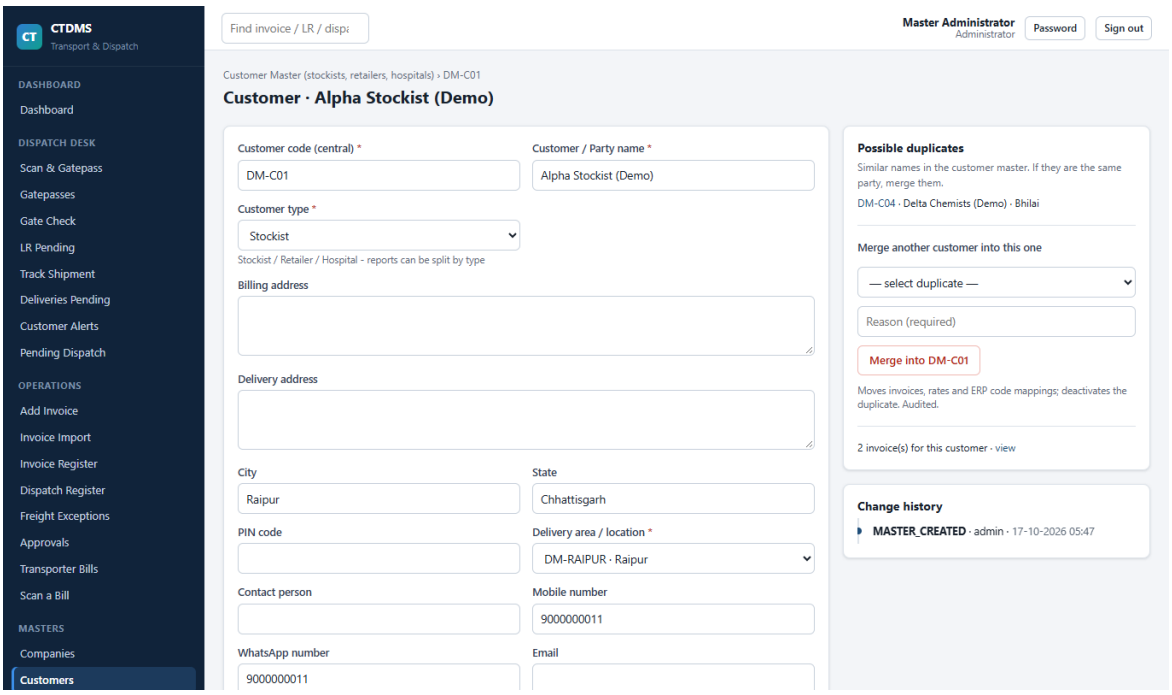
Companies, each tied to its billing system.



Customers: stockists, retailers and a hospital.

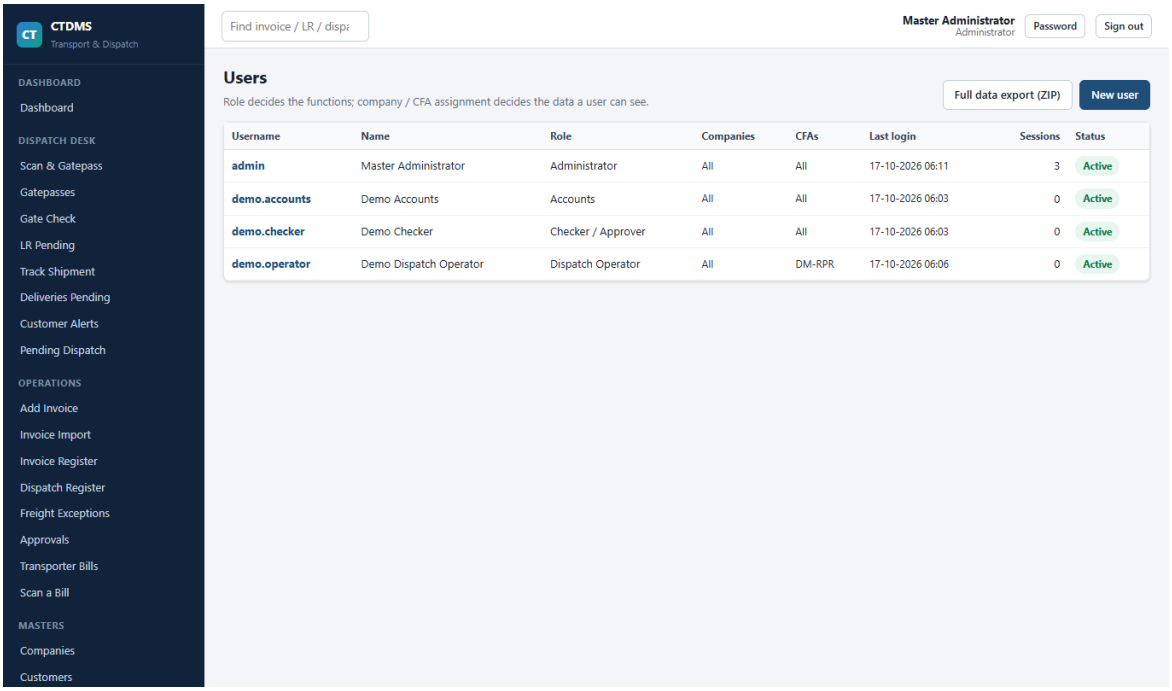


Transporters with payment mode.

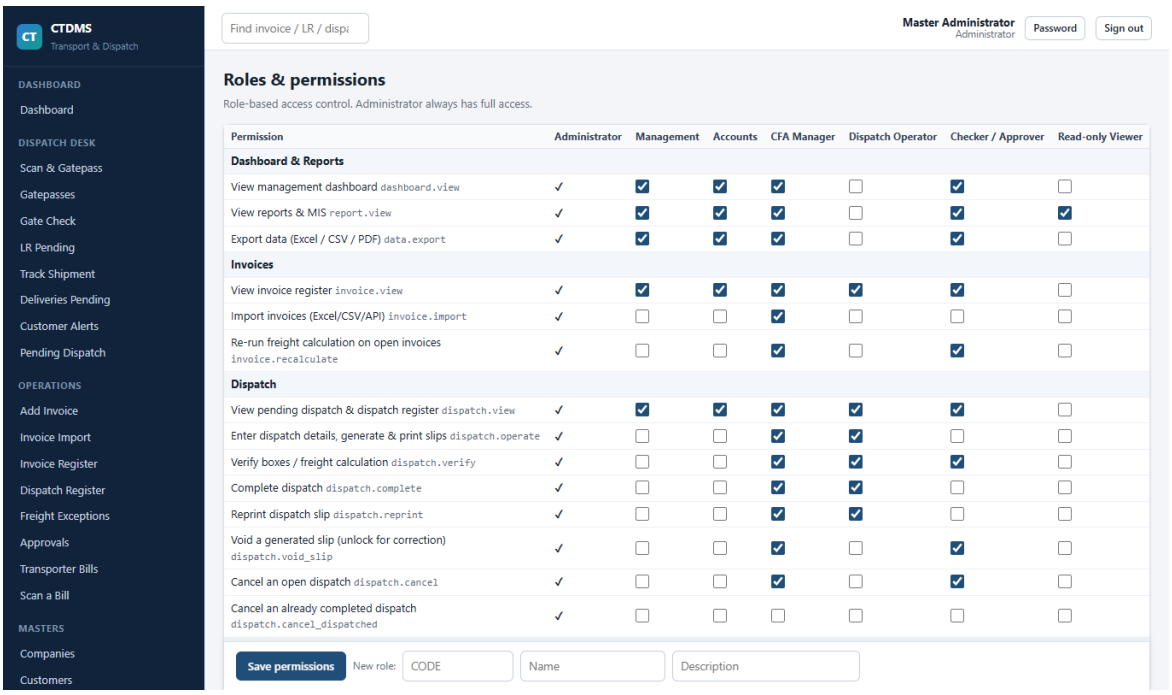


On a customer, record what each ERP calls it ('Add ERP code'); duplicates can be merged.

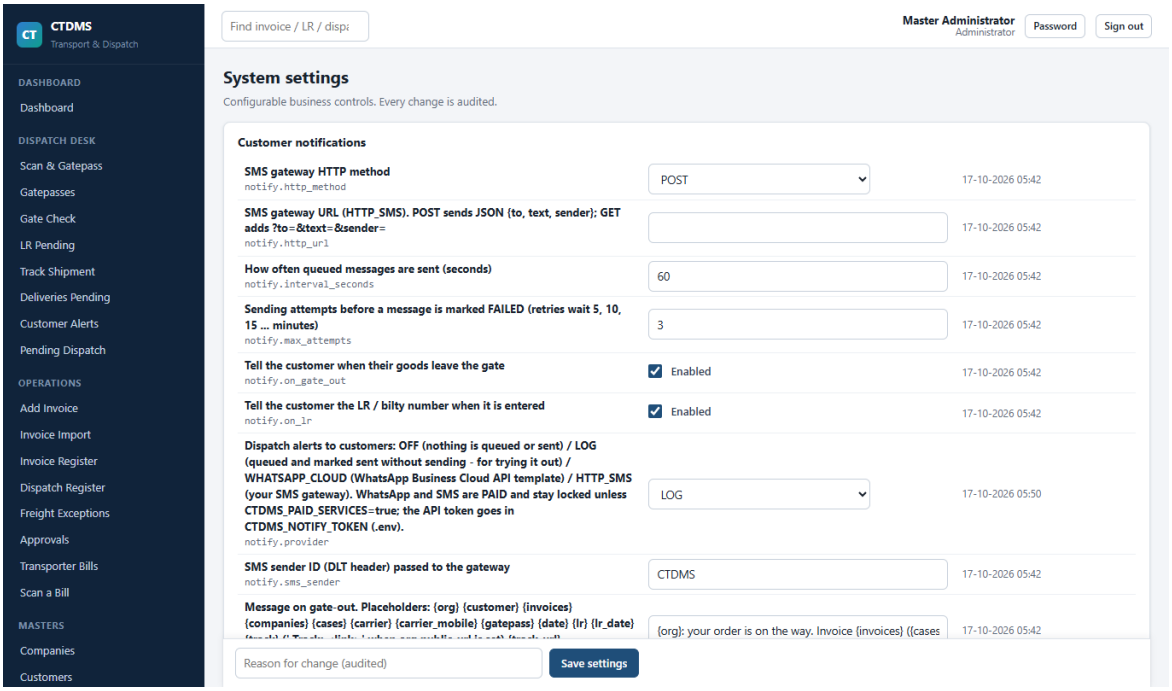
Step 4. Users, roles, settings



Users with role and last login.



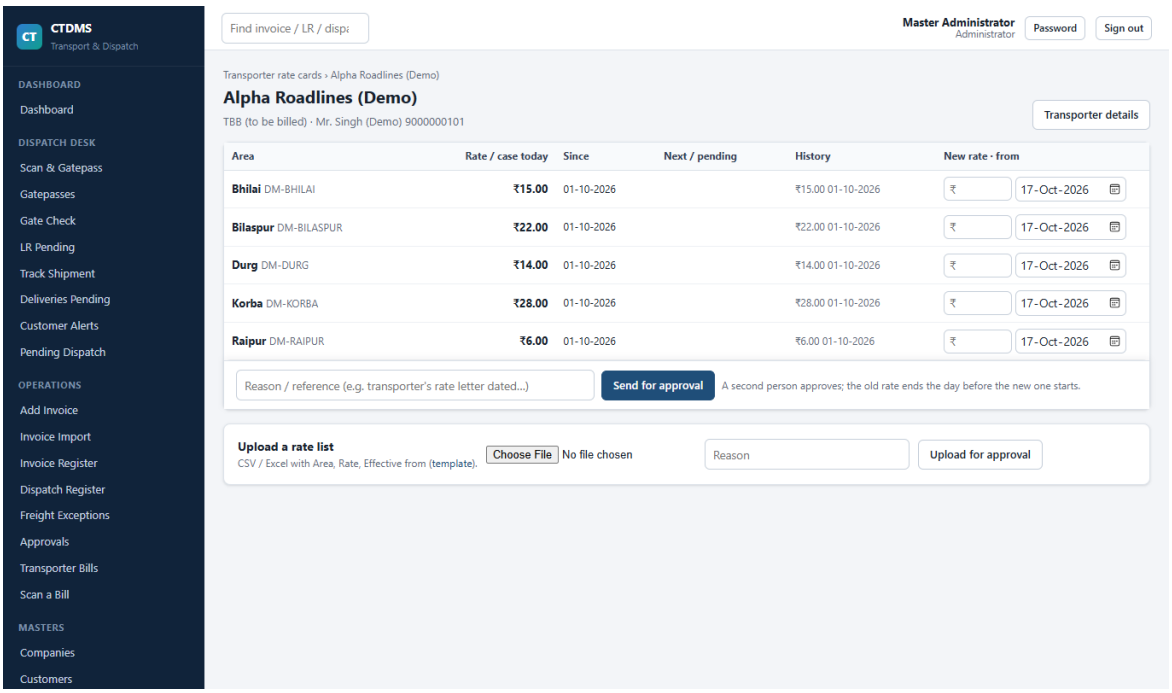
Roles and permissions grid.



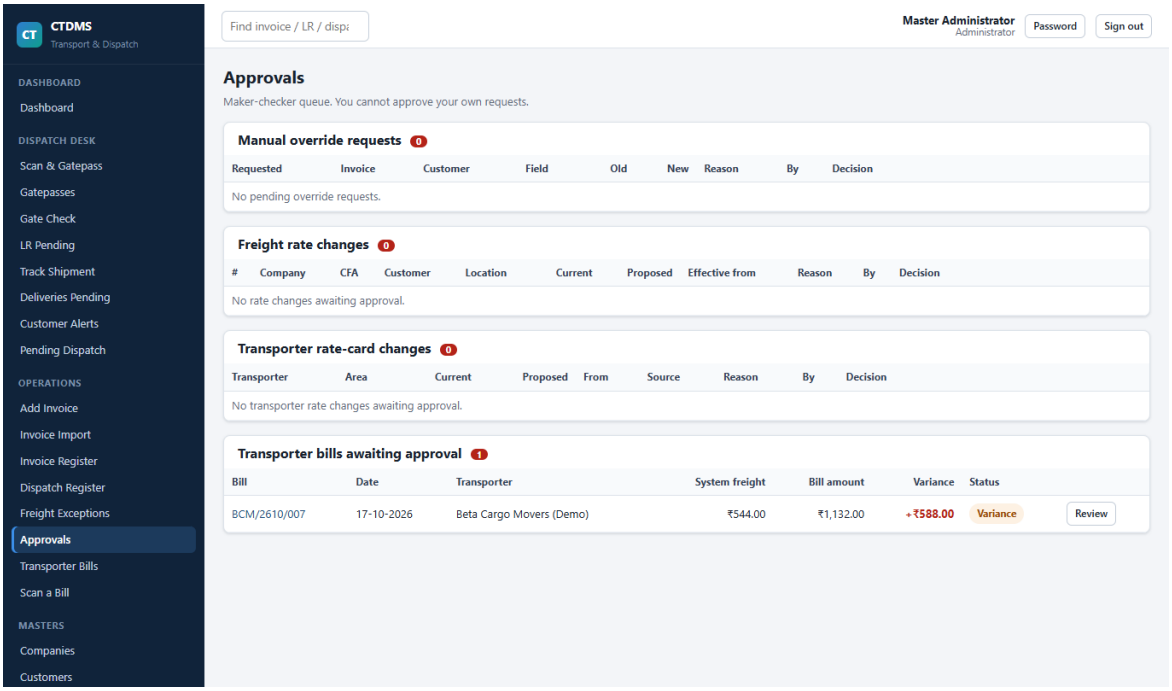
System settings: organisation name, alerts, dispatch rules, approvals, reconciliation tolerances.

Step 5. Rate cards and the two-person rule

Open a transporter in Masters › Transporter Rate Cards; upload a CSV (Area, Rate, Effective from) or type a rate and an effective date. Every rate goes to Approvals; a different person (the checker) presses Approve. Only approved rates price gatepasses.

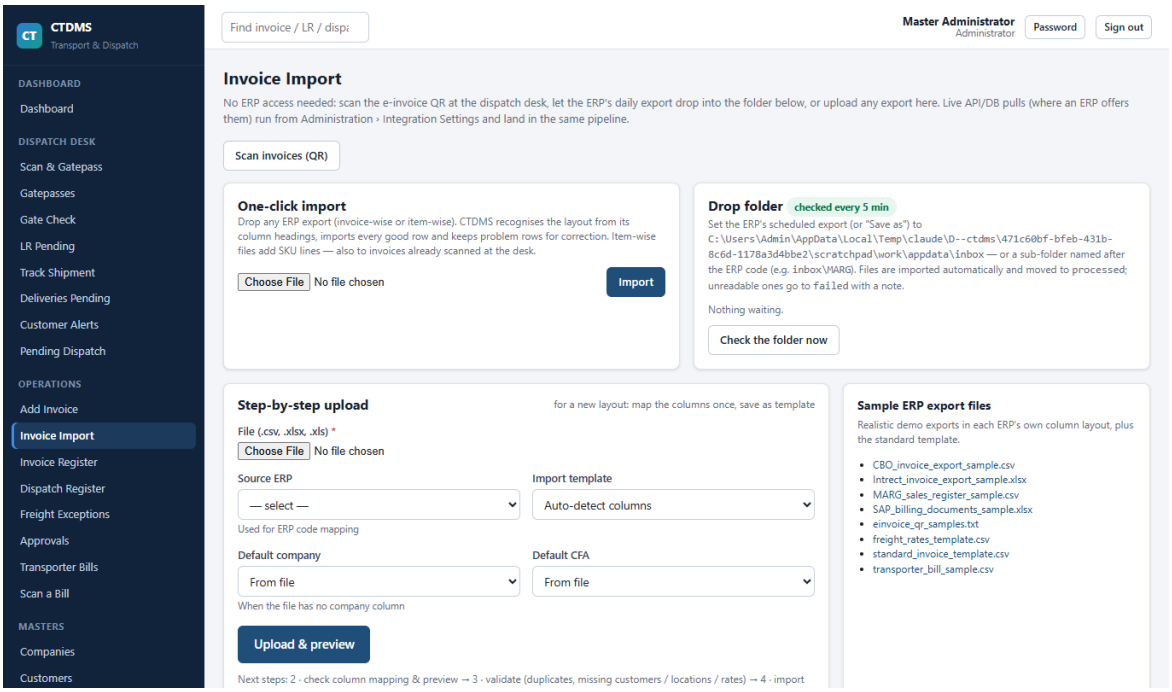


Alpha Roadlines' approved rate card.

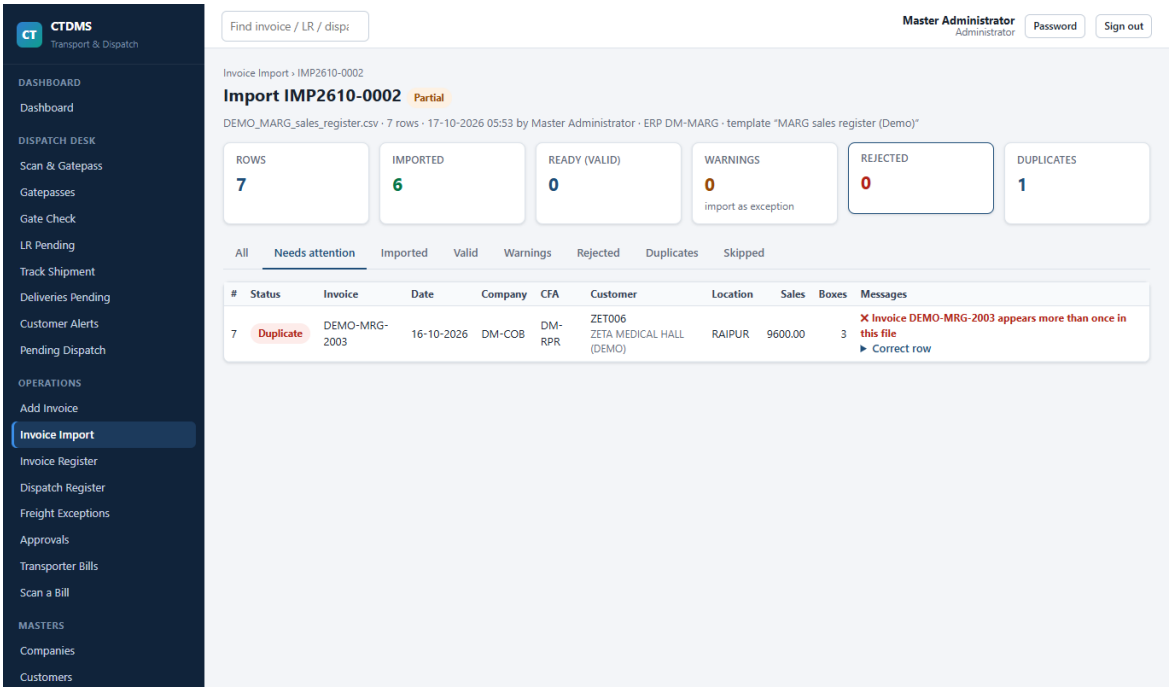


Approvals queue (empty once everything has been approved).

Step 6. Bring in the invoices

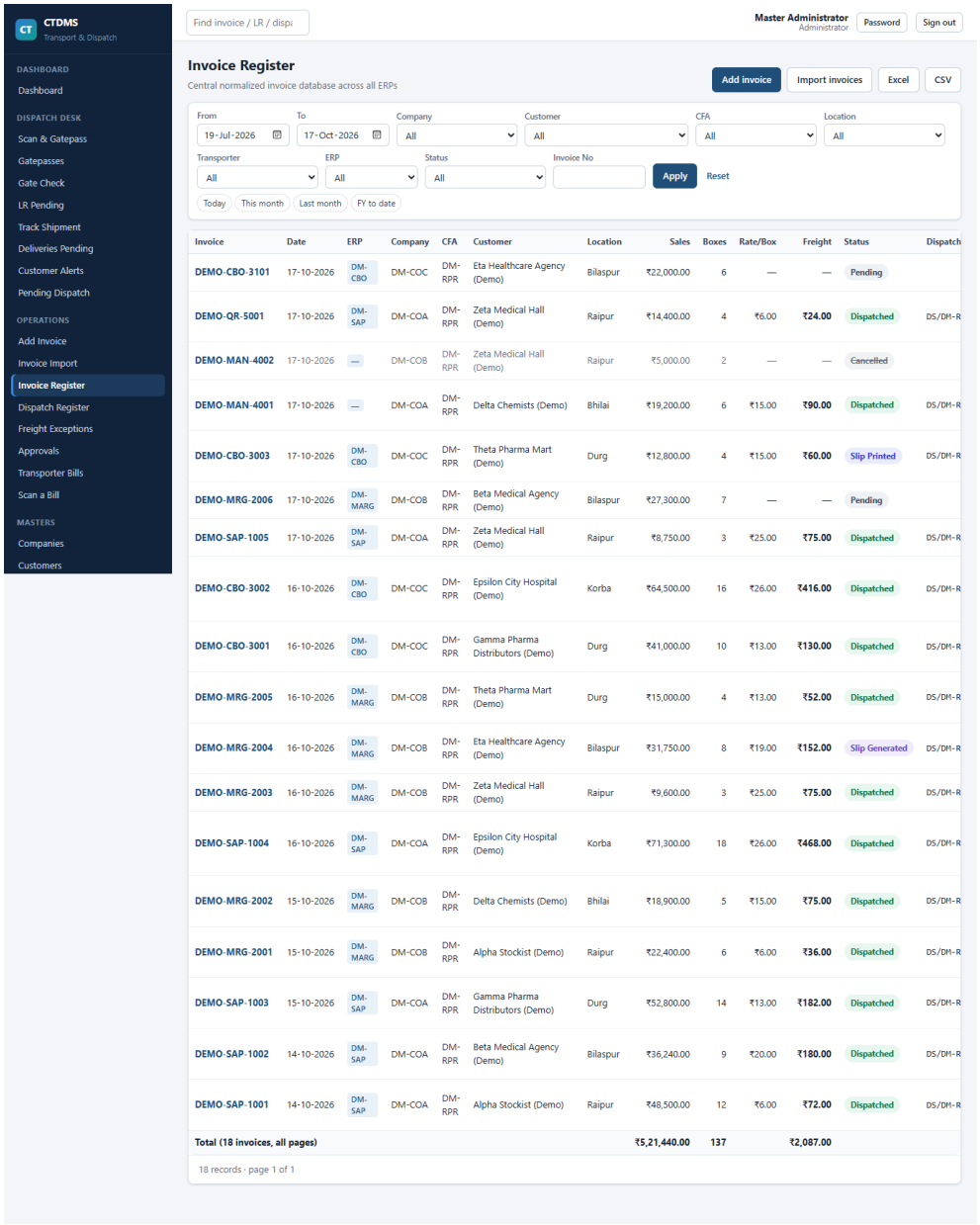


Invoice Import: one-click, step-by-step, drop folder and the import log.

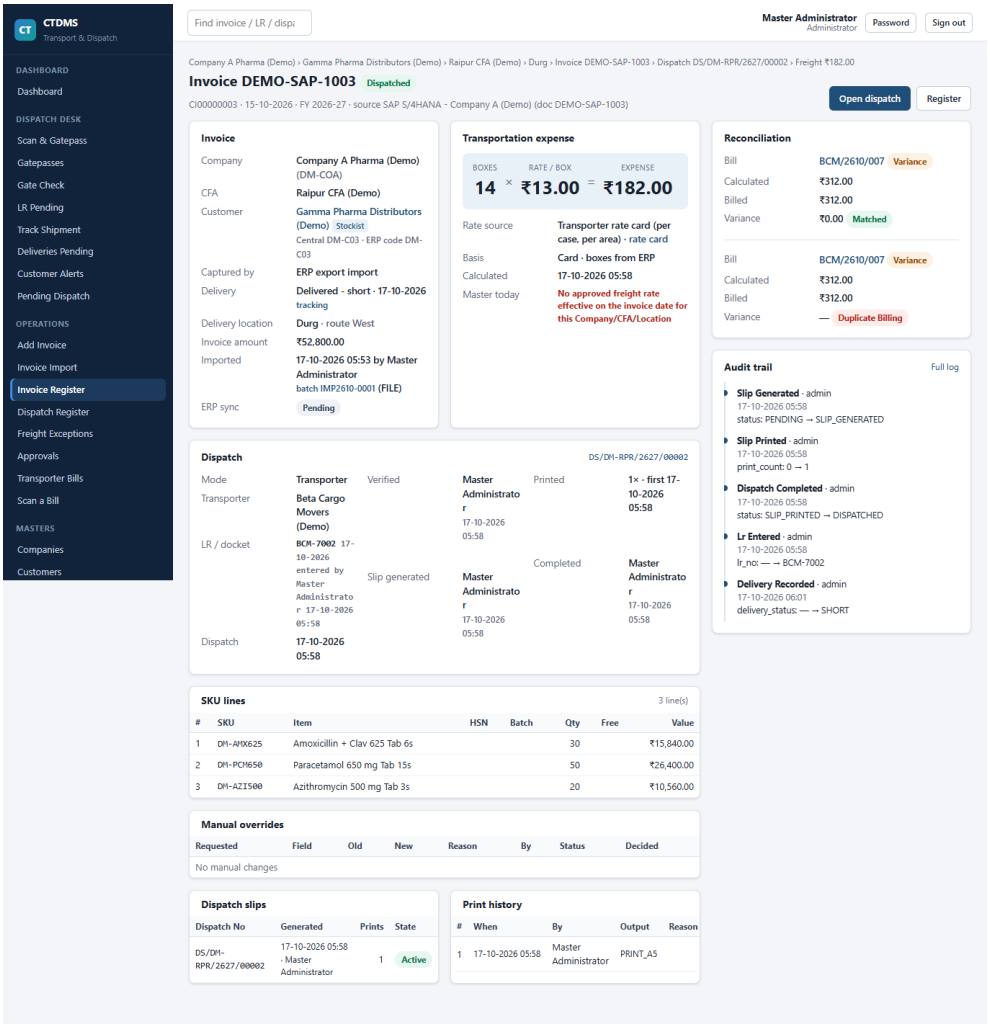


MARG file with three problems explained row by row: unknown party code (with a suggested match), no case count, duplicate.

- Choose the ERP, company and CFA, press **Upload & preview**; CTDMS reads the column headings.
- Fix what is flagged: *Map code*, *Correct row*, *Skip*. Name the layout so next time it is recognised automatically.
- Press **Import N valid rows**. Nothing is stored before you press it.

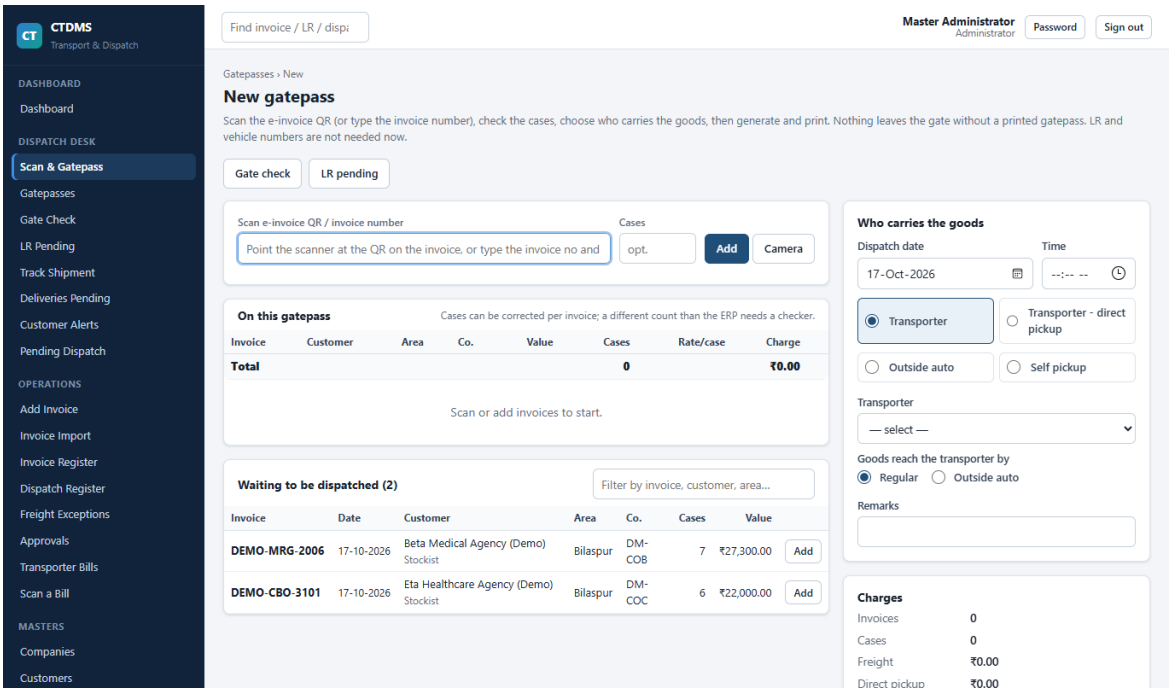


Invoice Register: all sources, all statuses, in one list.

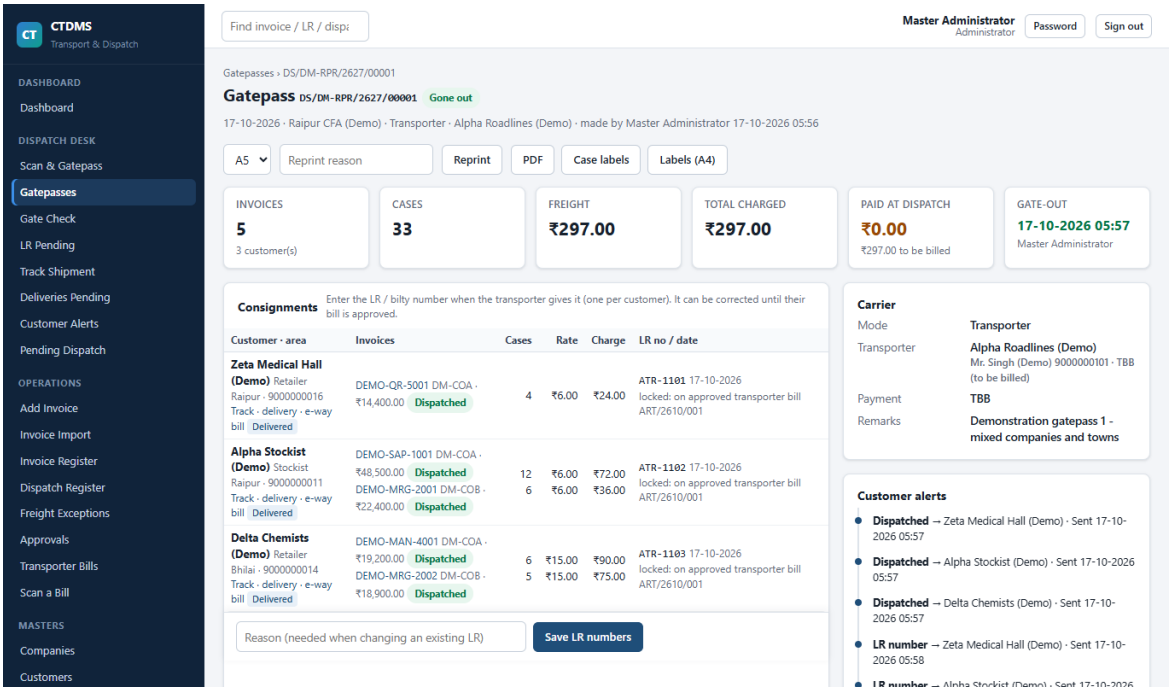


One invoice in full: freight calculation, dispatch, SKU lines, overrides, prints, reconciliation and audit trail.

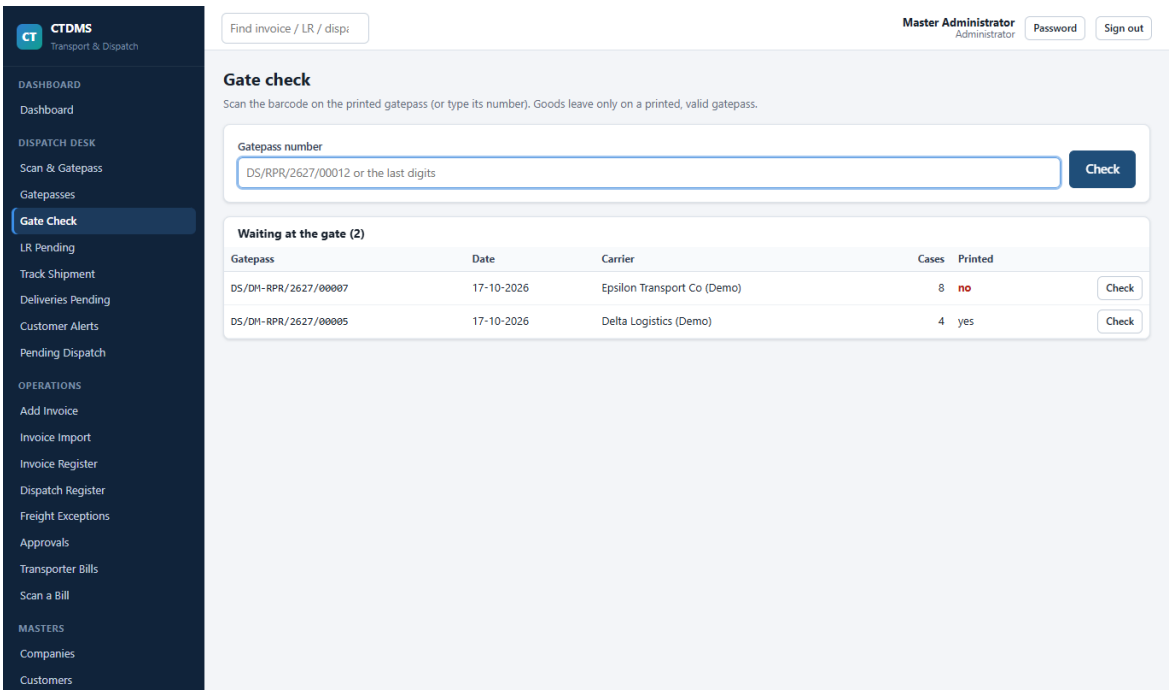
Step 7. The gatepass: scan, group, print, gate-out, LR



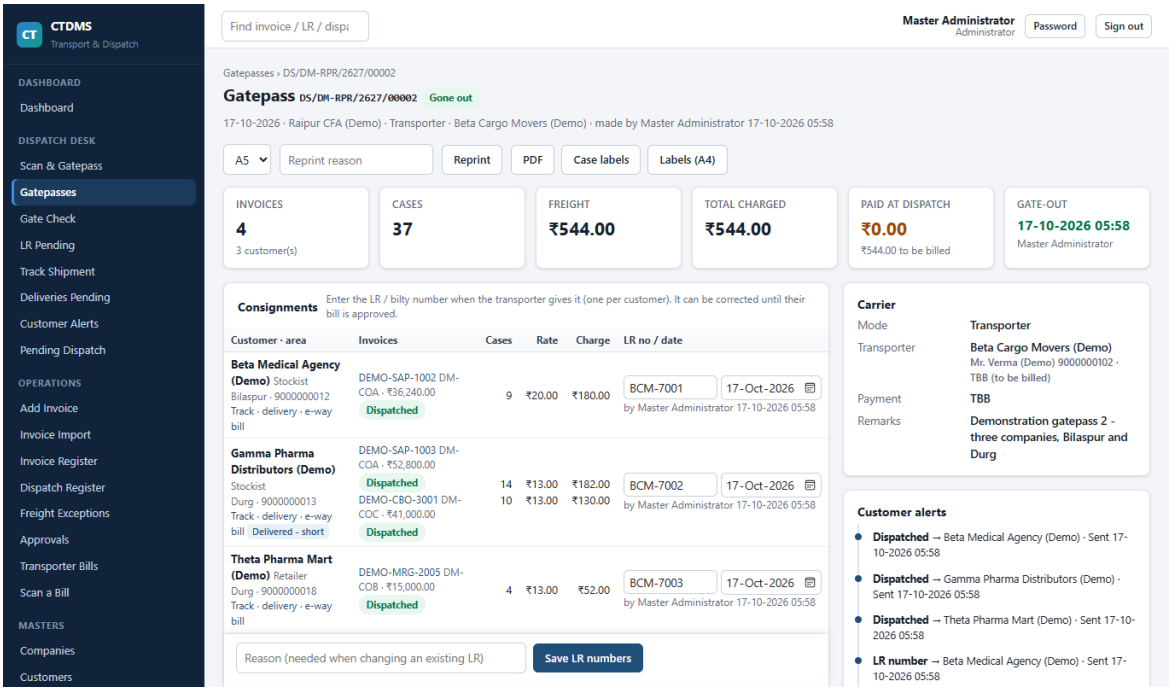
Scan & Gatepass: scan the QR or press Add on waiting invoices; choose the carrier; see the charges.



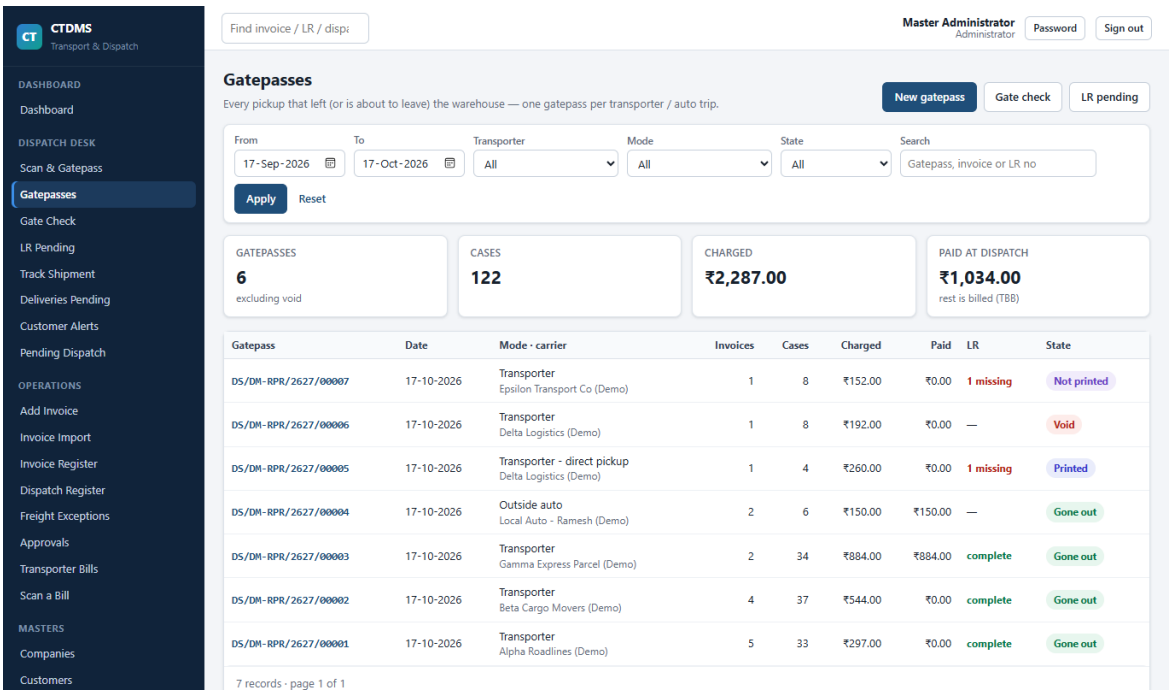
Gatepass 1 after gate-out: five invoices, three customers, two towns, LR numbers entered.



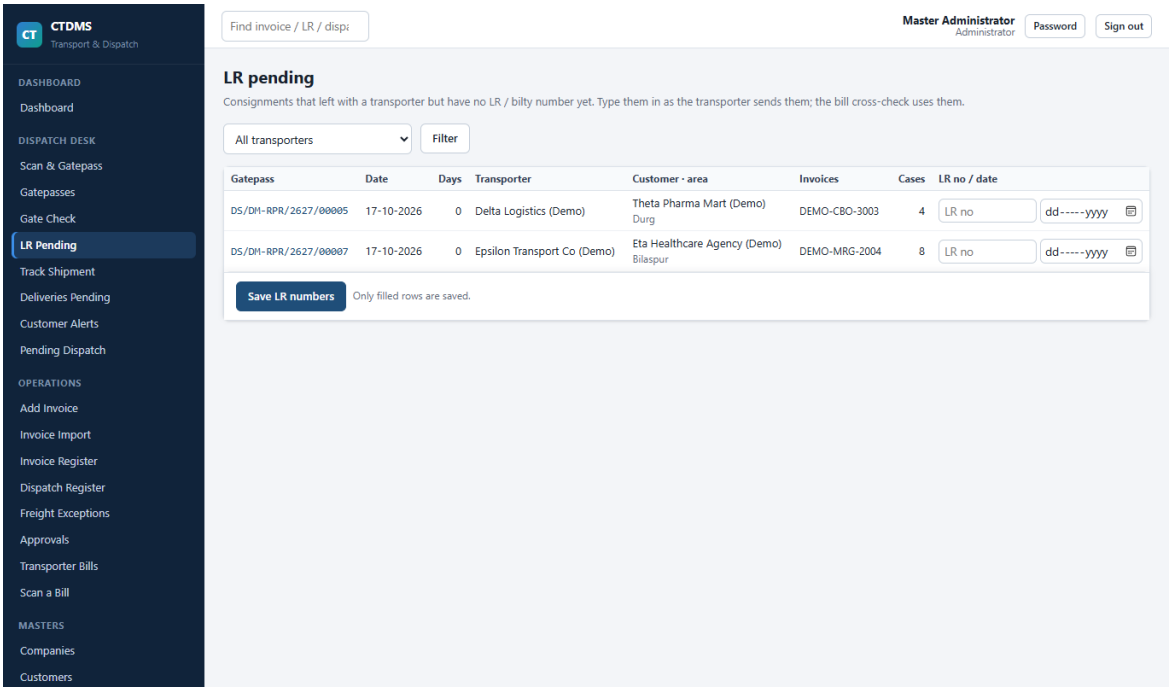
Gate Check: the guard types or scans the gatepass number.



Gatepass 2: three companies in one consignment.

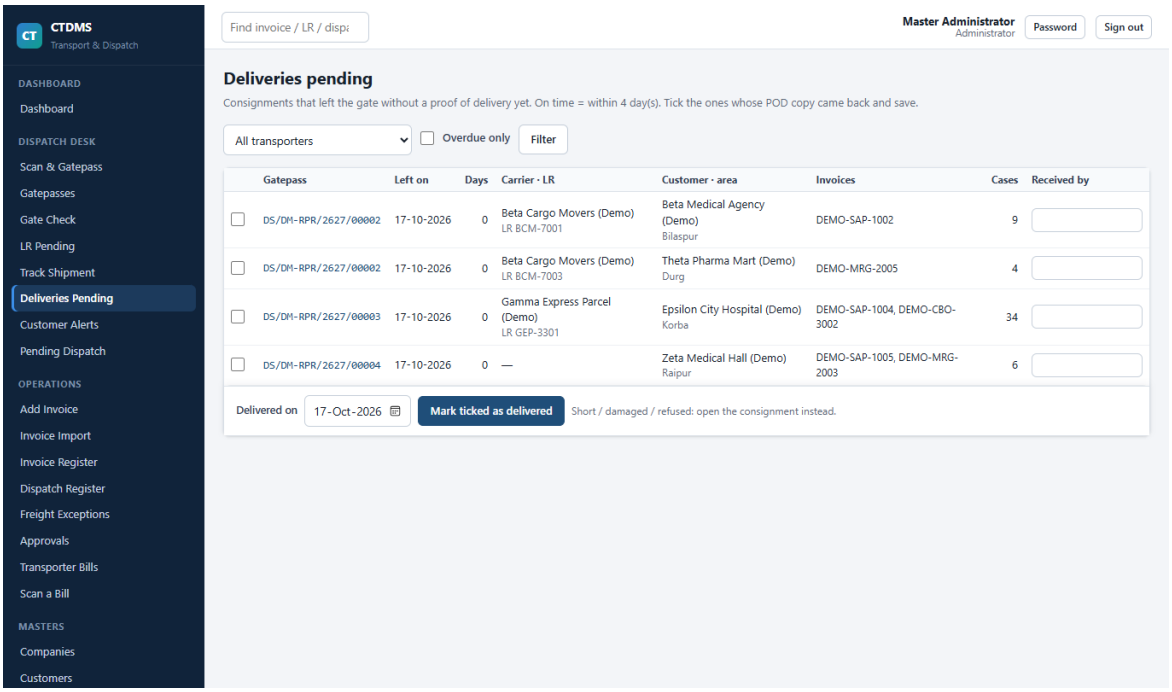


The gatepass register: gone out, printed, not printed, void.

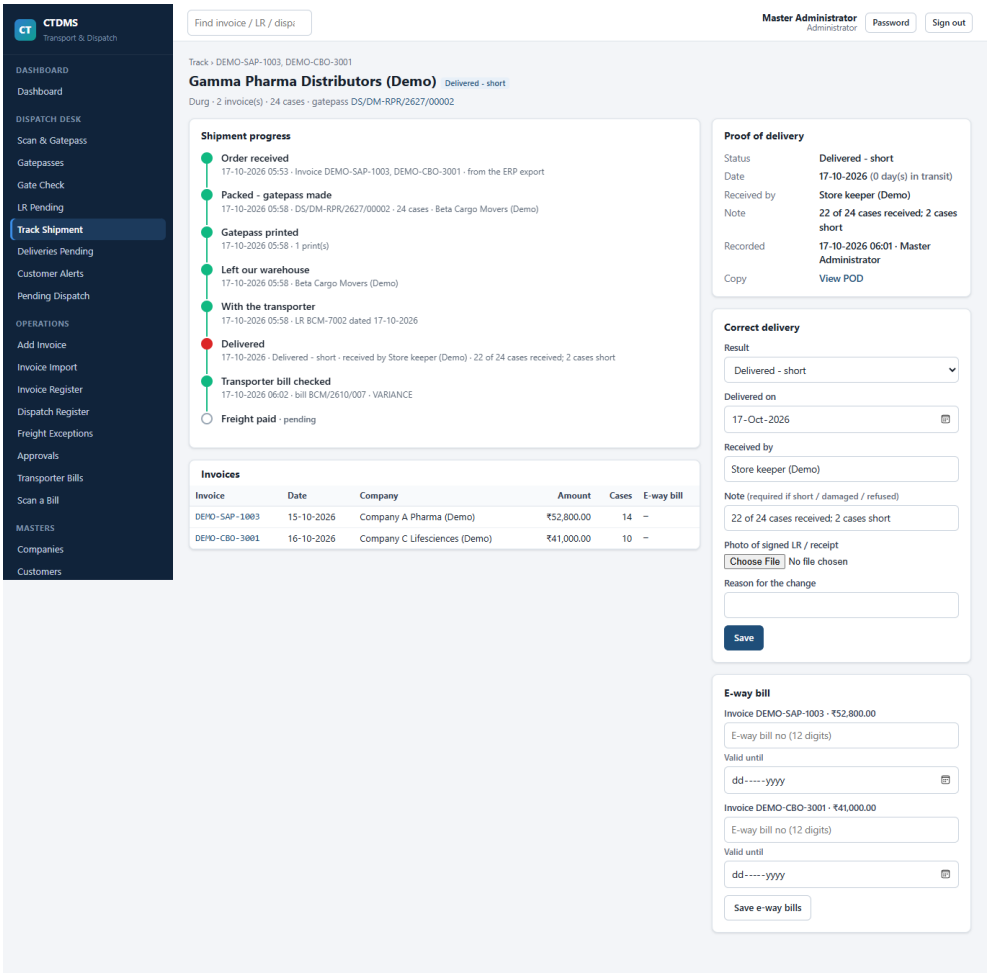


LR Pending: consignments still waiting for the transporter's LR number.

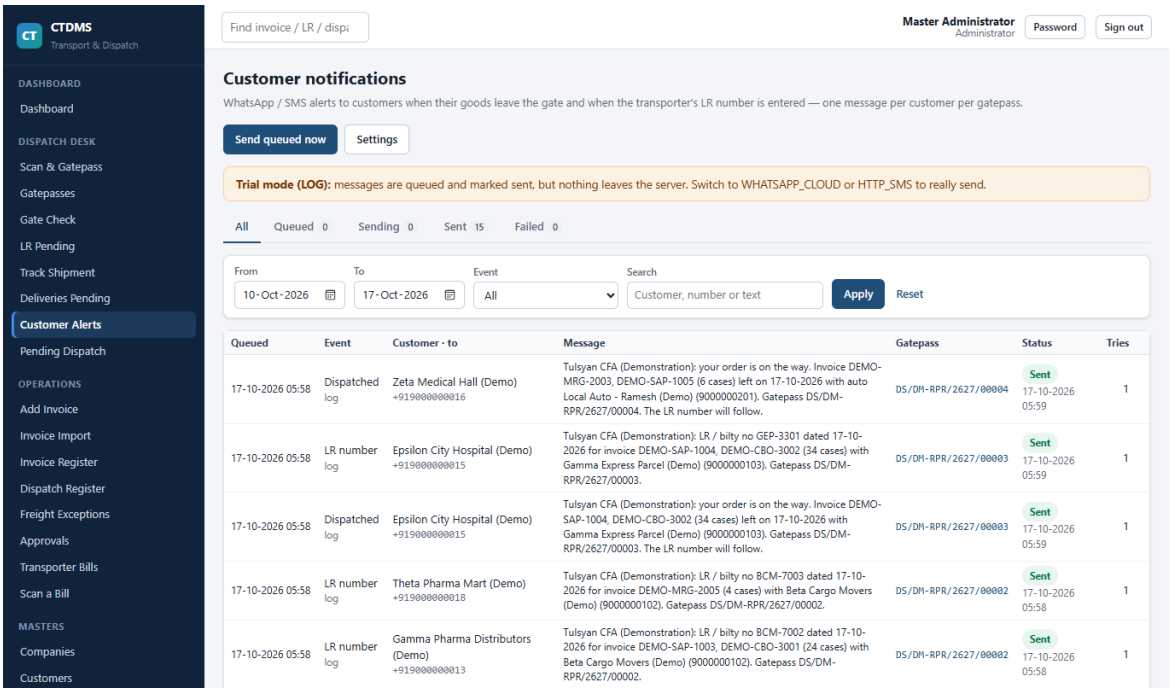
Step 8. Delivery, proof and tracking



Deliveries Pending (tick, choose the date, mark delivered).

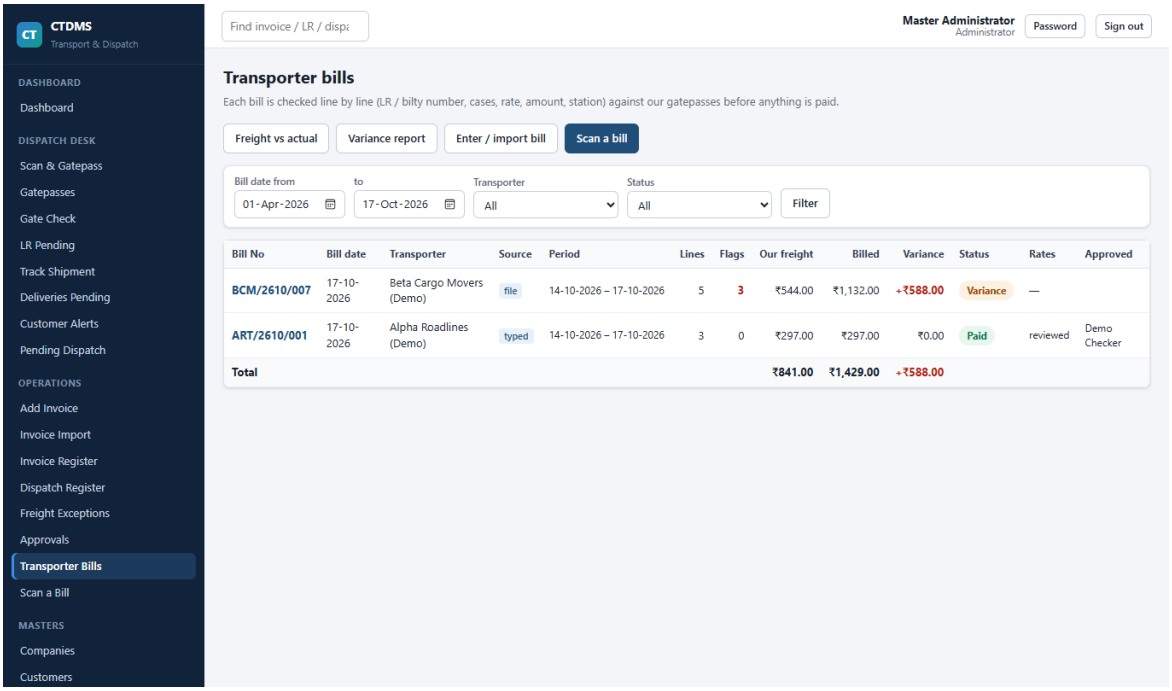


Track Shipment: the full timeline of one consignment, with proof of delivery recorded as short.

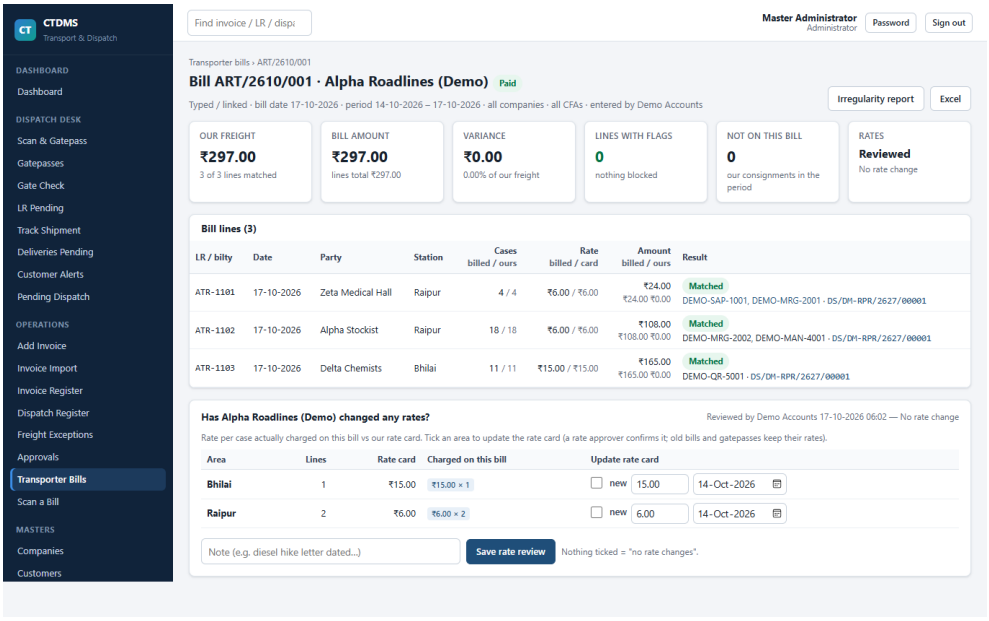


Customer Alerts in preview mode.

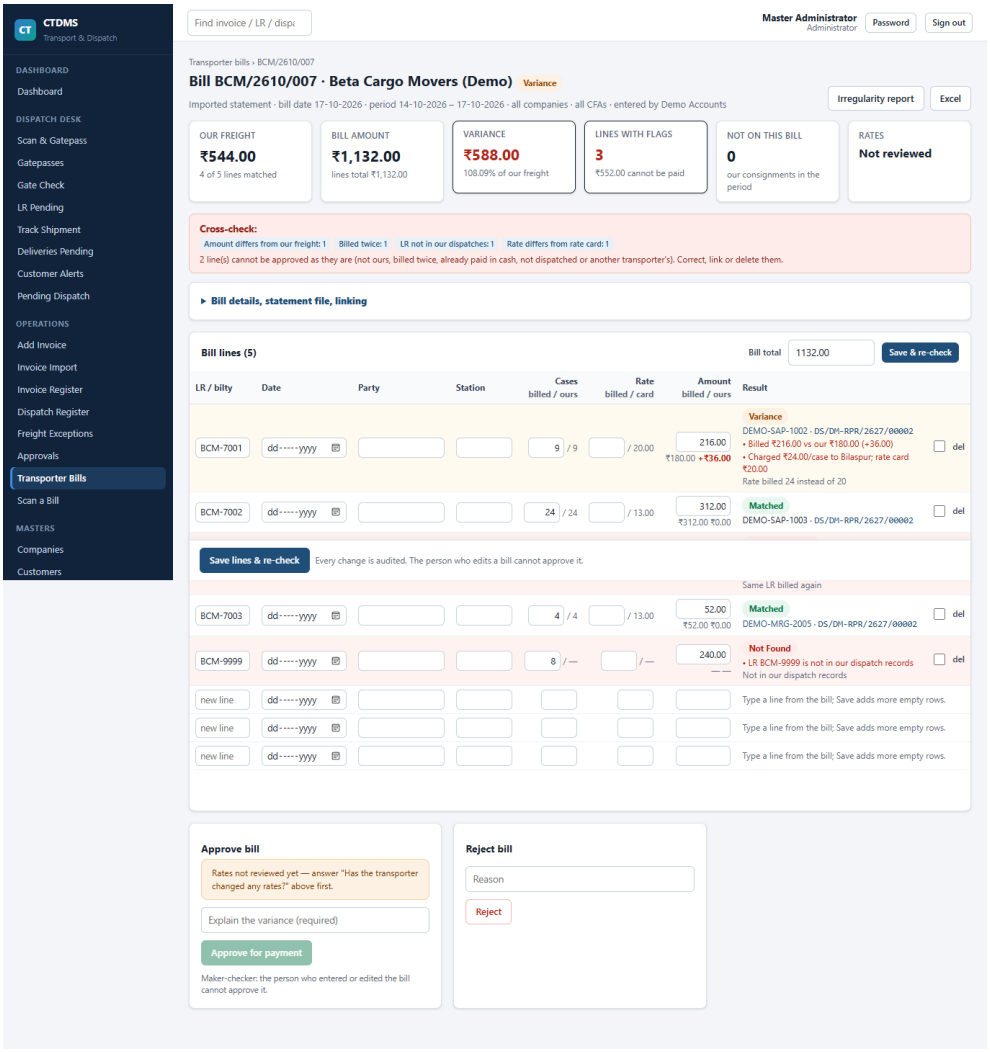
Step 9. Transporter bills



Transporter bills and their status.

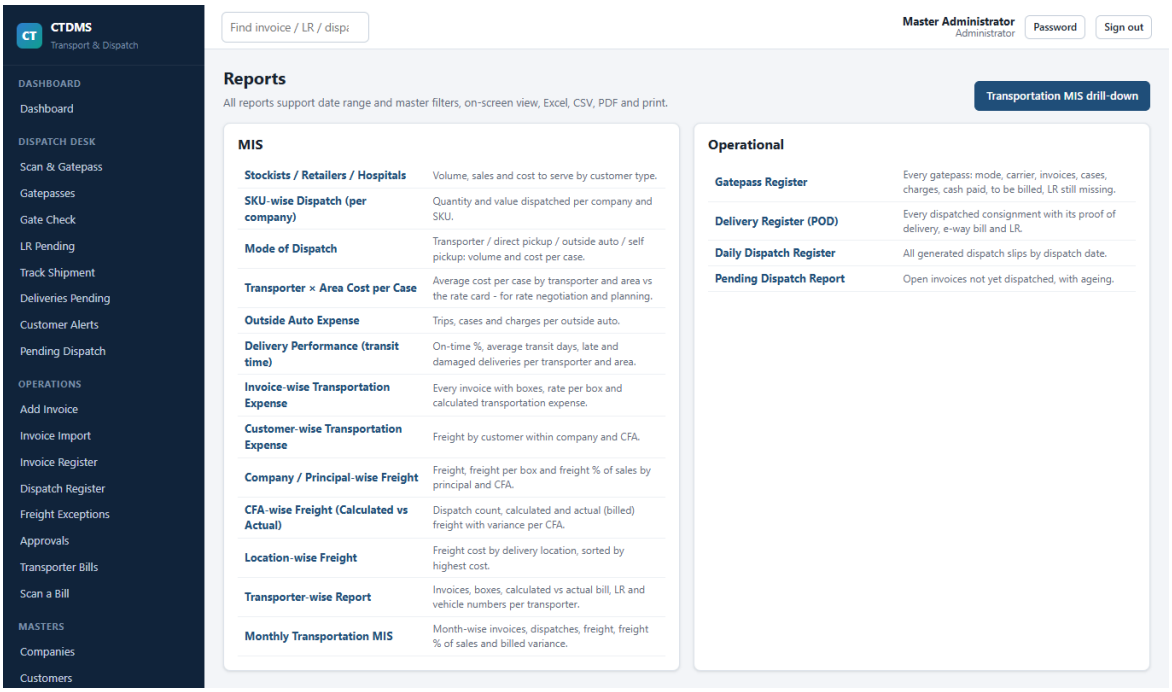


Bill 1: all lines matched, variance zero, approved by the checker and paid.

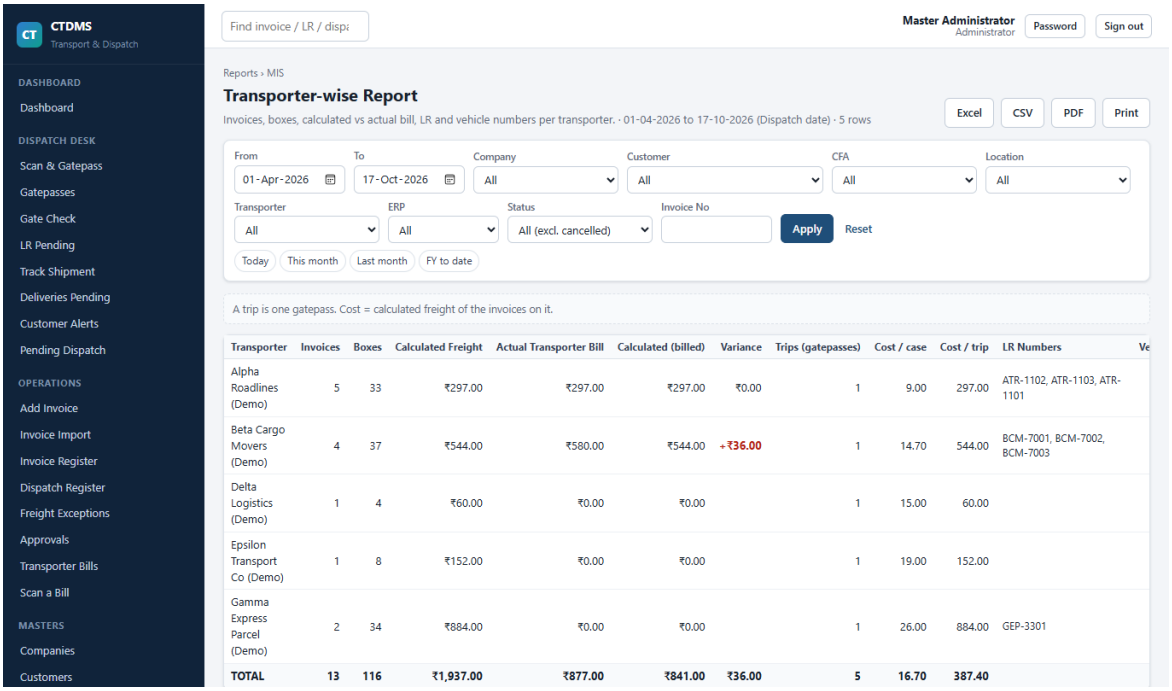


Bill 2: five flagged lines; it cannot be approved.

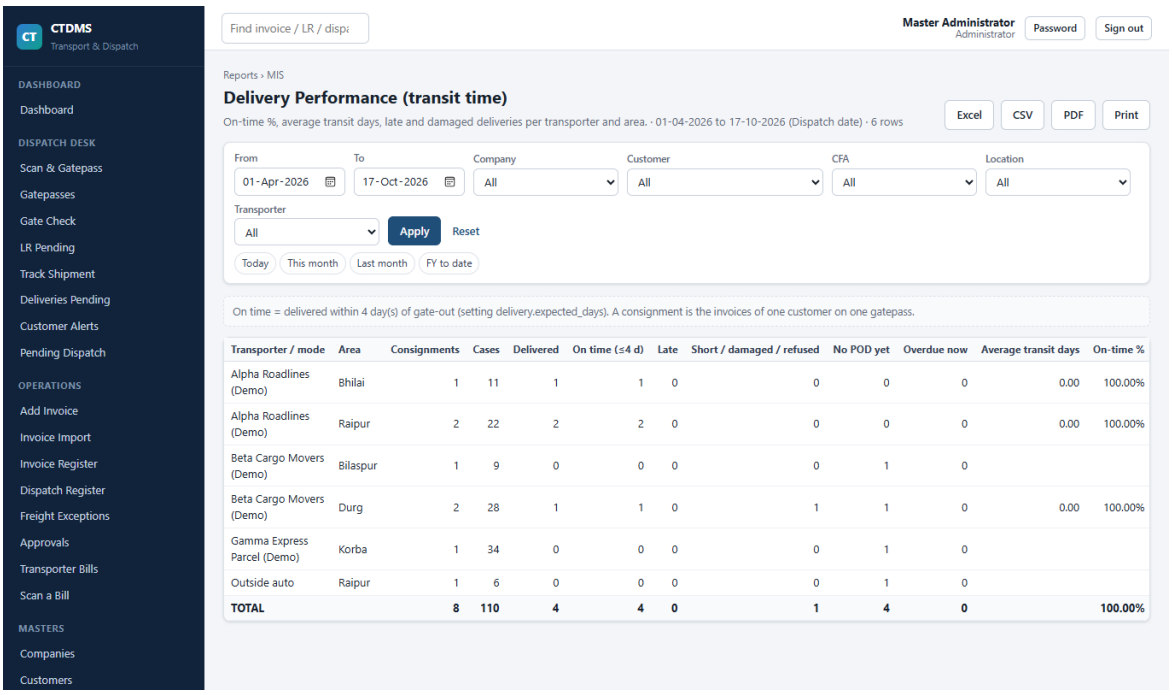
Step 10. Reports and exports



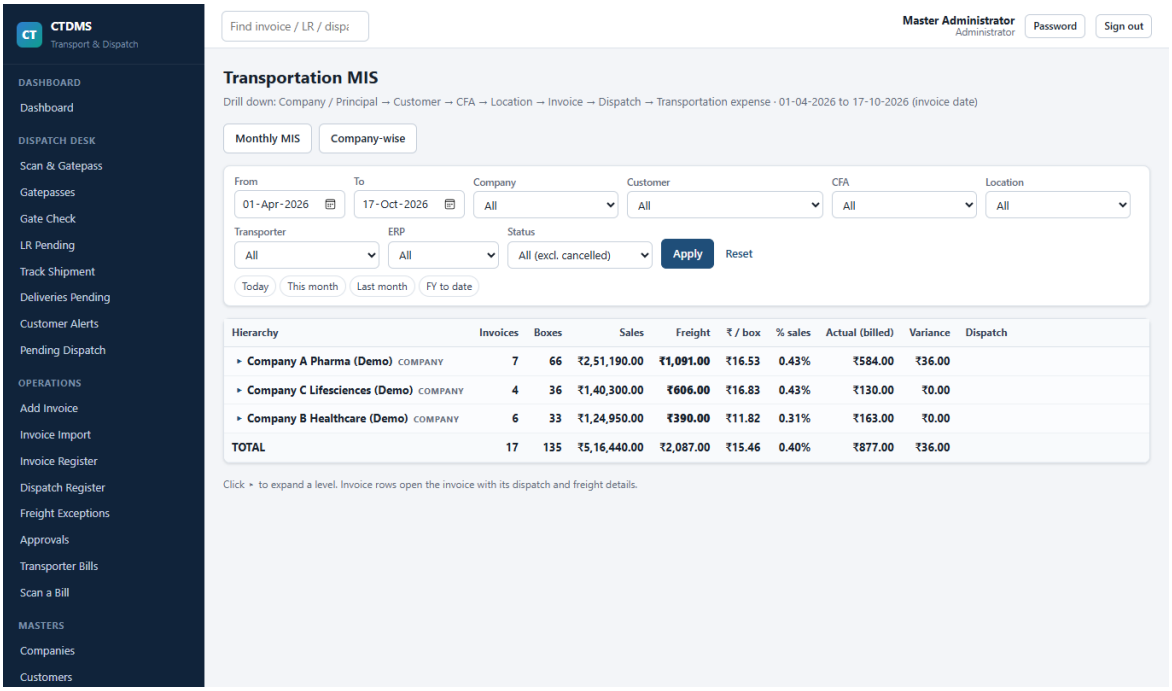
The full list of reports.



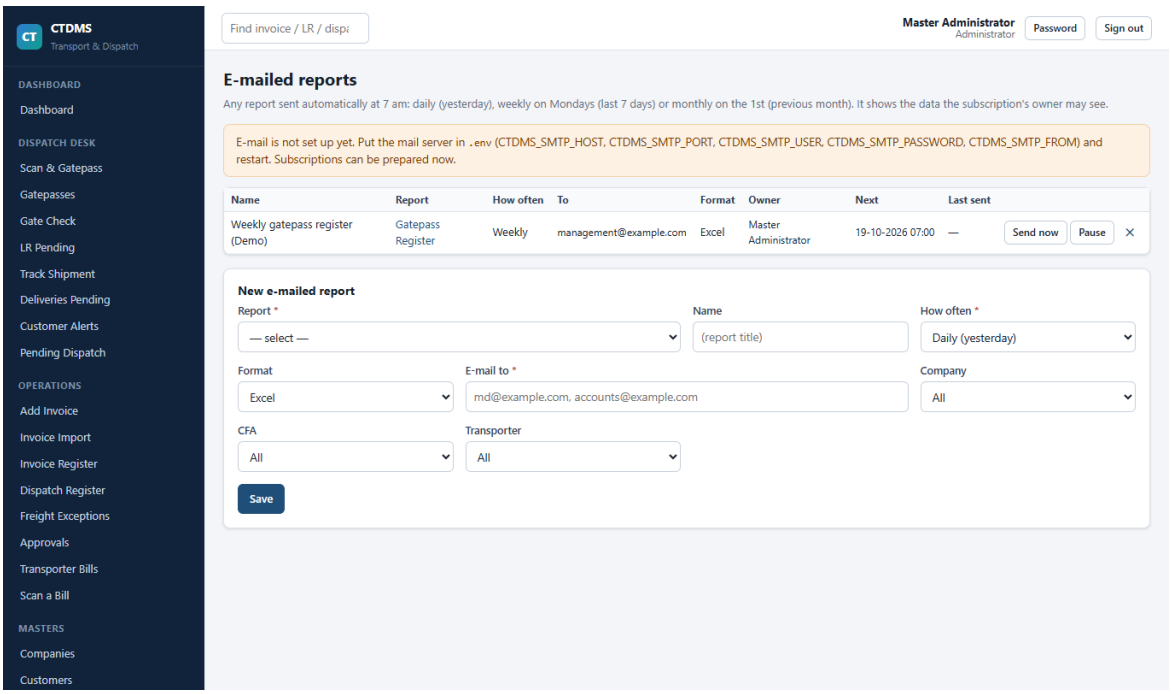
Transporter-wise: trips, cases, cost per case, cost per trip; export buttons at the top.



Delivery performance by transporter and area.

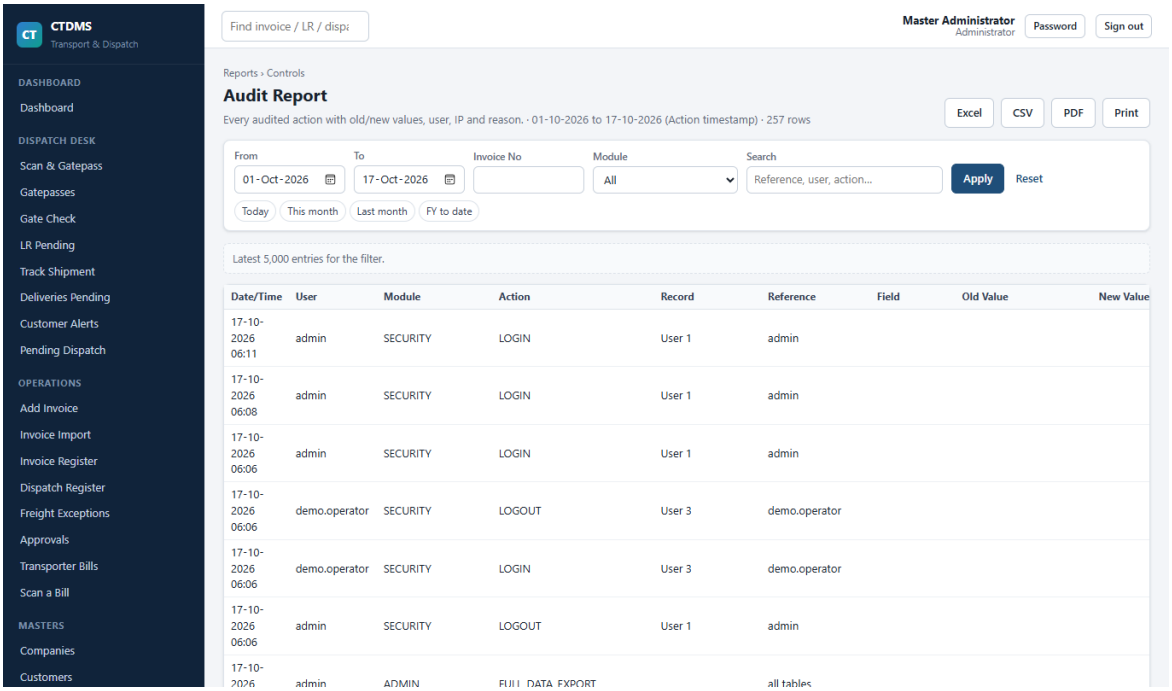


Transportation MIS.



E-mailed reports.

Step 11. Control: audit



Audit log: who changed what and when.

7. All the formats in which data can be taken out and used

Each format below was produced during the demonstration. The sample files are in the **exports** folder.

Format	How to get it	Good for	Demonstration file(s)
Printed gatepass (A5 / A4)	Gatepass page › Print gatepass	Paper copy carried with the goods, barcode for the gate	(printed screen in the video)
PDF gatepass	Gatepass page › PDF	E-mail to the transporter, archive	Gatepass_G1_print.pdf
Case labels (100 × 50 mm or A4)	Gatepass page › Case labels	One label per case, with barcode	Gatepass_G1_case_labels.html
Excel report (.xlsx)	Any report › Excel	Management analysis, pivot tables	Report_Gatepass_Register.xlsx, Report_SKU-wise_Dispatch.xlsx, Report_Delivery_Register_POD.xlsx
CSV report (.csv)	Any report › CSV	Import into other software	Report_Stockists_Retailers_Hospitals.csv, Report_Transporter-wise.csv
PDF report (.pdf)	Any report › PDF; Audit log › PDF	Printing, board papers	Report_Audit_Log.pdf, Report_Transporter-wise.pdf
Irregularity report for a bill	Bill page › Irregularity report / Excel	Discussion with the transporter	Report_Transporter_Bill_Irregularities.xlsx
E-mailed reports	Reports › E-mailed Reports (daily / weekly / monthly)	Management receives reports automatically	(subscription saved; needs mail server)
Full data export (.zip of CSVs)	Administration › Users › Full data export	A complete copy of the data for review or migration	Full_data_export.zip
REST API (JSON)	/api/v1/... with a token; documentation at /api/docs	Other software reads invoices, tracking, reports, dashboard	API_dashboard_summary.json, API_invoices.json, API_push_result_CBO.json, API_report_transporter.json, API_track_BCM-7002.json
API write-in (ERP push)	POST /api/v1/integrations/{ERP}/invoices with the ERP's API key	ERP sends invoices without files	API_push_result_CBO.json
Drop-folder import	ERP saves its export into data/inbox; read every 5 minutes	Automatic daily file import	(described in the user guide; not run in this demonstration)
Customer tracking link	Alerts contain a signed link when org.public_url is set	Customers see the timeline without prices	(needs public address and paid alerts)
Screen (live)	Dashboard, MIS drill-down, Track Shipment	Day-to-day viewing	the video

8. Problems that appeared (on purpose) and what to do

The demonstration contains these real-life problems. Each one is visible in the video.

Problem shown	What CTDMS did	What the person does
MARG party code 'TH-08' unknown	Rejected the row, suggested the closest customer	Press 'Map code'; never needed again
MARG invoice without a case count	Imported as an Exception; cannot be dispatched	Enter the cases (override, audited); other users need approval
Invoice twice in one file	Marked Duplicate, not stored twice	Skip it
Customer code unknown in CBO file	Matched through the GST number	Nothing
Transporter has no rate for the area	Shows 'no rate' beside the invoice at the gatepass	Choose another transporter, or add the rate and get it approved
Gatepass made on the wrong transporter	Void with a reason; invoices return to the waiting list	Make a new gatepass
Gatepass not printed	Gate-out is refused	Print, then confirm gate-out
Short delivery	Recorded on the consignment with note and photo	Follow up with the transporter; appears in Delivery Performance
Invoice entered by mistake	Cancel with a reason; stays in history	None
Rate billed higher than the rate card	Flagged on the bill line	Query the transporter, or approve with remarks if agreed
LR billed twice	Blocks approval	Delete the duplicate line
LR not in our records	Blocks approval; may suggest a likely consignment	Link it to the right consignment or remove it
Person tries to approve own work	Refused (two-person rule)	Ask the checker
Wrong password five times	Account locked for 15 minutes	Master: Users › Set password & unlock

9. Before real use

1. Create a fresh, empty database: *python -m app.cli init-db*, then *python -m app.cli create-admin --username … --password …*. Do not copy anything from the demonstration database.
2. Enter your real ERPs, CFAs, areas, companies, customers, transporters and rate cards, in the order of section 3. Ask a second person to approve each rate card.
3. Create each real user with the right role and scope. Remove or deactivate the demonstration users if they were ever created.
4. Set the organisation name and address, and review the dispatch rules in System Settings.
5. Use the drop folder or file import first; add API connections later.
6. Keep paid services (AI bill reading, WhatsApp, SMS) locked unless the business decides to pay for them.
7. Make a database backup and test a restore.

CAUTION Demonstration logins (for the demonstration database only): admin / Master@2026; demo.checker, demo.operator, demo.accounts / Demo@Pass2026. Change them if the demonstration is ever shared outside the office.

Starting the demonstration again

Double-click **start_demonstration.bat** in this folder. It starts the application on **http://127.0.0.1:8002** with the demonstration database. Close the window to stop. Your real data is not involved.